



UNIVERSITY OF THE PELOPONNESE & NCSR "DEMOCRITOS"
MSC PROGRAMME IN DATA SCIENCE

**Use of Deep Learning methods and land
cover/use data to improve the spatial resolution
of Numerical Weather Prediction (NWP)
simulations.**

by

Antigoni Moira

A thesis submitted in partial fulfillment
of the requirements for the MSc
in Data Science

Supervisor: Theodoros Giannakopoulos
Principal Researcher, NCSR "Demokritos"

Co-supervisors: Stylianos Karozis
Principal Researcher, NCSR "Demokritos"

Athens, July 2025

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Antigoni Moira

MSc. Thesis, MSc. Programme in Data Science

University of the Peloponnese & NCSR “Democritos”, July 2025

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Athens, July 2025

Acknowledgments

I would like to express my sincere gratitude to my supervisor, Dr. Theodoros Giannakopoulos, for the opportunity to undertake this thesis and for his guidance throughout the process.

I am especially thankful to my co-supervisor, Dr. Stelios Karozis, whose invaluable support, consistent feedback, and domain insight played a crucial role in shaping this work. I am deeply grateful for the time and expertise he generously shared with me.

I would also like to thank my colleague, Mr. Nikos Gounaris, for providing the land use and elevation data. Finally, I would like to express my heartfelt gratitude to the important people in my life, whose quiet encouragement and support have helped me stay motivated throughout this journey.

Περίληψη

Σκοπός της παρούσας εργασίας είναι η υλοποίηση μιας μεθόδου βασισμένης σε τεχνικές Βαθιάς Μάθησης για τη βελτίωση της χωρικής ανάλυσης των αποτελεσμάτων των προσομοιώσεων αριθμητικών προγνωστικών μοντέλων καιρού. Συγκεκριμένα, επιλέχθηκε να μελετηθεί η θερμοκρασία σε ύψος δύο μέτρων από την επιφάνεια της θάλασσας, καθώς αποτελεί παράμετρο-κλειδί για τομείς όπως η γεωργία και η υγεία, ενώ παράλληλα διαθέτει χαρακτηριστικά που διευκολύνουν την ανάλυση της (είναι συνεχής μεταβλητή με μικρότερη χωρική διακύμανση σε σύγκριση με άλλες παραμέτρους, όπως ο άνεμος).

Για τη διαμόρφωση της κατάλληλης μεθοδολογίας και την επιλογή της βέλτιστης αρχιτεκτονικής των μοντέλων Βαθιάς Μάθησης που χρησιμοποιήθηκαν, προηγήθηκε έρευνα της διαθέσιμης βιβλιογραφίας καθώς και δοκιμές εναλλακτικών αρχιτεκτονικών. Αρχικά, στα δεδομένα εφαρμόστηκαν κλασικές τεχνικές αναδειγματοληψίας (resampling), ώστε τα αποτελέσματά τους να χρησιμοποιηθούν ως σημείο αναφοράς (baseline) για την αξιολόγηση της απόδοσης των μοντέλων Βαθιάς Μάθησης. Στη συνέχεια, δοκιμάστηκαν και συγκρίθηκαν οι επιδόσεις των ακόλουθων μοντέλων, τα οποία χρησιμοποιούνται συχνά για τη βελτίωση της χωρικής ανάλυσης εικόνων (Super Resolution): Convolutional Auto-Encoder (CAE), Super-Resolution Convolutional Neural Network (SRCNN), Enhanced Deep Super-Resolution Network (EDSR) και Residual Channel Attention Network (RCAN).

Παράλληλα, εξετάστηκαν διαφορετικές προσεγγίσεις μεθοδολογίας, οι οποίες αξιοποιούν υπάρχουσες παρατηρήσεις και γνώσεις από το πεδίο των προγνωστικών μοντέλων. Αρχικά, εφαρμόστηκε σταδιακή αύξηση της χωρικής ανάλυσης του πλέγματος των δεδομένων από $100\text{km} \times 100\text{km}$ σε $50\text{km} \times 50\text{km}$ και στη συνέχεια σε $25\text{km} \times$

25km. Με αυτήν τη στρατηγική, το πρόβλημα που καλούνται να «μάθουν» να επιλύουν τα μοντέλα γίνεται πιο εύκολο, επιτρέποντας τη χρήση λιγότερο σύνθετων αρχιτεκτονικών και, συνεπώς, μειώνοντας τις υπολογιστικές απαιτήσεις. Επιπλέον, έγινε προσπάθεια ανάπτυξης εποχικών μοντέλων, λαμβάνοντας υπόψη ότι οι παράγοντες που επηρεάζουν τη θερμοκρασία μεταβάλλονται σημαντικά κατά τη διάρκεια του έτους και από εποχή σε εποχή. Τέλος, για τη βελτίωση της ευκρίνειας των αποτελεσμάτων υψηλής ανάλυσης, εξετάστηκε η δυνατότητα αξιοποίησης δεδομένων που σχετίζονται με τη μορφολογία του εδάφους και το υψόμετρο.

Abstract

The aim of this thesis is to implement a method based on Deep Learning (DL) techniques to improve the spatial resolution of the results obtained from numerical weather prediction models. Specifically, the study focuses on air temperatures 2 meters (T2m) above sea level, as it is a key parameter for sectors such as agriculture and public health.

To develop an appropriate methodology and select the optimal architecture for the DL models used, an extensive review of the available literature was performed, followed by testing alternative architectures. Initially, traditional resampling techniques were applied to the data and their results were then used as a baseline to evaluate the performance of the DL models. Subsequently, the performance of the following models, which are commonly used for Super Resolution in images, was tested and compared: Convolutional Auto-Encoder (CAE), Super-Resolution Convolutional Neural Network (SRCNN), Enhanced Deep Super-Resolution Network (EDSR) and Residual Channel Attention Network (RCAN).

In addition, different approaches were examined, taking advantage of existing observations and domain knowledge from numerical weather prediction models. Initially, a progressive increase in the spatial resolution of the data grid was applied, transitioning from $100\text{km} \times 100\text{km}$ to $50\text{km} \times 50\text{km}$ and then to $25\text{km} \times 25\text{km}$. This approach simplifies the learning task for the models, allowing the selection of shallower architectures, thus reducing computational costs. Furthermore, seasonal models were developed, taking into account the significant variations in temperature-driving factors from season to season. Finally, to further improve the clarity of HR results, the utilization of data related to terrain and elevation was explored.

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List of Abbreviations

DL	Deep Learning
NN	Neural Network
CAE	Convolutional Auto-Encoder
SRCNN	Super-Resolution Convolutional Neural Network
EDSR	Enhanced Deep Super-Resolution Network
RCAN	Residual Channel Attention Network
AI	Artificial Intelligence
ML	Machine Learning
CNN	Convolutional Neural Network
RCM	Regional Climate Model
GCM	Global Climate Model
MAE	Mean Absolute Error
MSE	Mean Squared Error
PSNR	Peak Signal-to-Noise Ratio
SGD	Structural Similarity Index Measure
SSIM	Stochastic Gradient Descent

LIST OF ABBREVIATIONS

ECMWF	European Centre for Medium-Range Weather Forecasts
T2m	2-meter air temperature
HR	High Resolution
MR	Medium Resolution
LR	Low Resolution
DEM	Digital Elevation Model
SISR	Single Image Super-Resolution

Chapter 1

Introduction

Weather and climate prediction have played a crucial role throughout human history. The earliest known attempts date back to 650 B.C., when the Babylonians sought to forecast short-term weather changes by observing cloud formations and optical phenomena like haloes. By around 340 B.C., the Greek philosopher Aristotle authored *Meteorologica*, a treatise exploring theories on the formation of rain, clouds, hail, wind, thunder, lightning, and hurricanes. Although Aristotle made several insightful observations, his work also contained notable inaccuracies. Nevertheless, his four-volume text remained the authoritative reference on weather theory for nearly 2,000 years. Many of his misconceptions persisted until the 17th century, when scientific advancements began to challenge and refine his ideas [10]. Today, more than two millennia later, meteorology has advanced exponentially, with cutting-edge technologies such as Remote Sensing and Artificial Intelligence (AI) enabling increasingly precise predictions with higher temporal and spatial resolution.

Weather prediction models can be classified based on either the geographical scale of the study area or the prediction timespan. Geographically, they are divided into (a) global and (b) regional forecast models. In terms of timespan, they fall into five categories: (i) nowcasting (hours), (ii) short-term forecasting (days), (iii) subseasonal forecasting (weeks), (iv) seasonal forecasting (months) and (v) climate projections (years) [11].

These models serve different purposes depending on the forecast timescale. Short-

term weather predictions (nowcasting and short-term forecasting) provide highly accurate and actionable insights for daily decision-making. These forecasts include key meteorological elements like temperature extremes, precipitation probability and intensity, wind characteristics, humidity levels, and cloud cover [12]. In contrast, medium-to-long-term climate predictions (subseasonal, seasonal forecasting and climate projections) aim to project broader climate patterns over months to years and offer insights into overarching climatic trends rather than immediate atmospheric conditions [1]. Examples of weather / climate prediction applications in daily life are summarized in Table 1.1.

Time Scale	Domains	Applications
Short Term	Agriculture	The timing for sowing and harvesting; Irrigation and fertilization plans.
	Energy	Predicts output for wind and solar energy.
	Transportation	Road traffic safety; Rail transport; Aviation and maritime industries.
	Construction	Project plans and timelines; Safe operations.
	Retail and Sales	Adjusts inventory based on weather forecasts.
	Tourism and Entertainment	Operations of outdoor activities and tourist attractions
	Environment and Disaster Management	Early warnings for floods, fires, and other natural disasters.
Medium — Long Term	Agriculture	Long-term land management and planning.
	Insurance	Preparations for future increases in types of disasters, such as floods and droughts.
	Real Estate	Assessment of future sea-level rise or other climate-related factors.
	Urban Planning	Water resource management.
	Tourism	Long-term investments and planning, such as deciding which regions may become popular tourist destinations in the future.
	Public Health	Long-term climate changes may impact the spread of diseases and cause damage to the infrastructure.

Table 1.1: Applications of Short term and medium-long term weather/climate forecasting in daily life [1].

Another categorization of weather forecasting models can be made based on the methodology they use:

1. **Numerical Weather Prediction (NWP) Models:** These use mathematical equations to simulate atmospheric processes, relying on real-time observations and supercomputers to solve complex equations.
2. **Statistical and Machine Learning Models:** These models use historical weather data and statistical relationships rather than physical equations.
3. **Hybrid Models:** These combine NWP and AI/statistical methods to improve accuracy.

Traditional NWP systems, while competitive, face challenges in forecasting rare events and managing model errors due to data volume, gaps, resource constraints, and collaboration issues between relevant agencies[13]. In the 1990s and 2000s, advancements in AI, particularly Machine Learning (ML) and DL, helped address these issues, improving forecast accuracy. Key developments after 2010, such as GPU processing, convolutional neural networks (CNNs), and online datasets, accelerated progress in weather analysis. These advancements have led to hybrid models that integrate ML/DL with NWP, enhancing forecasting efficiency, accuracy, and adaptability. AI offers a distinct advantage over traditional NWP by learning complex patterns from large datasets, making it effective for post-processing, uncertainty quantification, and predicting extreme weather events [4]. However, the evaluation of these methods and the transition to "End to End AI Models" is still a work in progress as shown in the Figure 1.1.

This work focuses on the spatial resolution of weather forecast models. Higher spatial resolution generally enhances the representation of small-scale processes and topographic features but comes with a significant computational cost [14, 15]. This challenge is particularly critical in long-term climate models, where there is a trade-off between the length of the forecast period and spatial resolution. However, it also affects short- and medium-term models when computational resources or time constraints limit resolution.

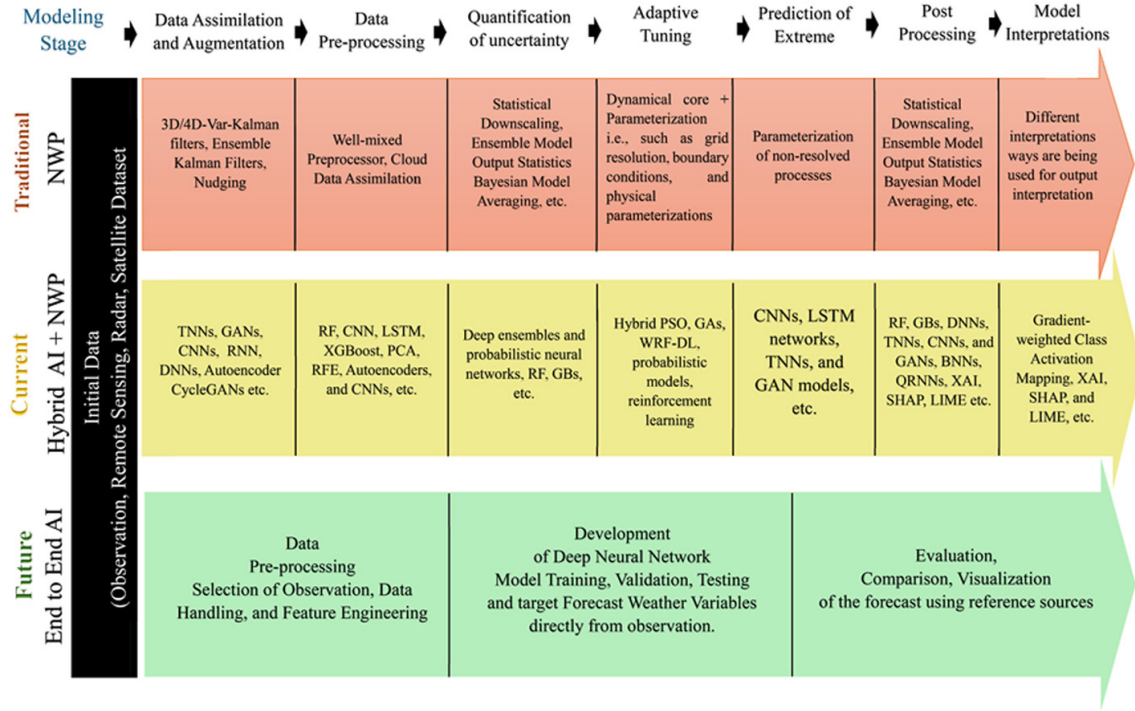


Figure 1.1: Artificial Intelligence in Weather Forecasting [4].

Regional Climate Modeling (RCM) is a powerful tool for overcoming the limitations of Global Climate Models (GCMs) by providing high-resolution climate data for specific regions [16]. By operating at finer spatial scales, RCMs offer an improved representation of local topography, land-sea contrasts, and extreme weather events [15]. This enhanced detail is particularly valuable in regions with complex terrain and for capturing short-lived climate phenomena (also known as short-lived climate pollutants such as black carbon, methane, tropospheric ozone, and hydrofluorocarbons) [16]. By refining our understanding of regional climate dynamics and variability, RCMs support GCMs and play a critical role in informing climate adaptation strategies for policymakers and stakeholders [17].

Recent studies have investigated the potential of deep learning DL techniques (Data-Driven Approaches) for downscaling global climate projections, comparing them with traditional RCMs (Physics-Based Approaches). DL-based downscaling offers significant advantages over RCMs in terms of computational efficiency and cost-effectiveness [18, 19]. The objective of this thesis is to implement and evaluate such a DL-based downscaling approach.

1.1 Problem description

This study addresses the challenge of enhancing the spatial resolution of climate data, focusing specifically on temperatures 2 meters (T2m) above sea level, which is a fundamental variable in climate analysis. The problem is formulated as a Single Image Super-Resolution (SISR) task, leveraging DL models to reconstruct HR temperature fields from their coarse-resolution counterparts.

The primary objective is to develop a downscaling approach that is agnostic to the source of input data (e.g. General Circulation Models (GCMs), weather models, or other reanalysis products) by training based on a single reanalysis dataset. LR inputs were generated through upscaling, ensuring consistency in data structure while avoiding the introduction of biases tied to specific model outputs.

In addition, domain knowledge was incorporated to improve model performance through enhancements like: multi-stage downscaling, season-specific models, and static geographic features integration such as elevation.

1.2 Thesis structure

The next chapter introduces key concepts related to SISR, outlines commonly used evaluation metrics, and provides essential definitions to support the understanding of the content in subsequent chapters.

Chapter 3 offers a comprehensive review of methodologies applied in SISR and climate data downscaling, with particular emphasis on recent DL approaches that have influenced the direction of this work.

Chapter 4 presents the DL models employed in this study, highlighting the architectural innovations and design choices specific to each.

Chapter 5 details the methodology used for data preprocessing, model training, and performance evaluation, presented clearly step by step.

Chapter 6 discusses the experimental results, explaining how each stage informed the next and evaluating model outputs based on both qualitative and quantitative criteria.

Finally, **Chapter 7** summarizes the key findings of this study and offers suggestions for future research directions and potential improvements.

Chapter 2

Theoretical Framework and Definitions

This chapter provides key definitions and outlines classical approaches relevant to the study, offering foundational knowledge to support the understanding of the concepts and methods presented in the subsequent chapters.

2.1 Key Definitions and Terminology

This subsection provides definitions for the Single Image Super-Resolution (SISR) problem and reanalysis data, which serves as the primary dataset used in this study to address the downscaling task.

2.1.1 Single Image Super-Resolution

Single Image Super-Resolution (SISR) is an image processing technique aimed at reconstructing a HR image from a single LR input [20, 21]. This distinguishes it from Multi-Image Super-Resolution (MISR), which utilizes multiple LR images of the same scene to enhance reconstruction quality by aggregating complementary information. SISR, on the other hand, relies solely on information from a single input image, making the task inherently more challenging as it involves inferring missing high-frequency details and mapping a low-dimensional representation to a higher-dimensional one, which is an ill-posed problem with potentially multiple valid

solutions[22].

Despite its complexity, SISR plays a crucial role in various domains such as remote sensing, medical imaging, biometric identification, security, and surveillance. Moreover, enhancing image resolution can significantly improve the performance of downstream computer vision tasks like object detection and semantic segmentation.

SISR methods can be broadly divided into two categories: traditional interpolation-based techniques and modern DL-based approaches [23]. Classical methods such as nearest-neighbor and bicubic interpolation have been widely used, while recent advances in deep learning have introduced more sophisticated and effective solutions, some of which are also explored in this study.

2.1.2 Reanalysis Data

Atmospheric reanalysis refers to the process of reconstructing past atmospheric conditions by integrating historical observational data (from in situ weather stations, weather balloons, aircrafts, ships, satellites, and other sources) with modern NWP models through data assimilation techniques. This approach leads to comprehensive, consistent, and spatially complete datasets that span extended periods, providing a detailed depiction of the Earth’s climate system over time [24].

In the era of green energy and climate change mitigation, reanalysis data plays an essential role in evaluating renewable energy availability, such as long-term patterns of solar irradiance or wind speed at specific locations. Moreover, reanalysis provides the observational estimates necessary to improve climate and weather models. These data serve as benchmarks for model validation and refinement, improving the precision of seasonal forecasts and climate projections [25]. They also provide consistent boundary conditions and initial states for driving other numerical models, such as HR regional climate models, hydrological simulations, or air quality models, ensuring that these downstream applications are grounded in a realistic representation of the large-scale atmosphere.

ERA5[3], ERA5-Land[26], and CERRA [27] are three of the most widely used reanalysis datasets provided by Copernicus:

1. ERA5 is the fifth-generation global atmospheric reanalysis produced by the ECMWF under the Copernicus Climate Change Service (C3S). It provides hourly data on a wide range of atmospheric, land-surface, and sea-state variables at a spatial resolution of 31 km ($0.25^\circ \times 0.25^\circ$), covering the period from 1940 to the present.
2. ERA5-Land is a high-resolution land component of ERA5, offering improved representation of land surface variables (e.g., soil moisture, temperature, snow cover) at 9 km ($0.1^\circ \times 0.1^\circ$) resolution. It uses the same land data assimilation system as ERA5 but is optimized for land applications and covers data from 1950 onward.
3. CERRA (Copernicus European Regional ReAnalysis) is a regional reanalysis focused on Europe, developed under Copernicus to provide very high-resolution (5.5 km) climate data for the European domain. It is designed to capture finer-scale phenomena such as topographic and coastal effects that global reanalyses like ERA5 may miss.

2.2 Methods and Approaches

This subsection outlines several traditional downscaling methods, some of which are employed in this study as baseline comparisons.

2.2.1 Image Interpolation

Interpolation is a fundamental technique in image processing, commonly used for operations such as zooming, resizing, rotating, and performing geometric corrections on digital images. It involves estimating pixel values at unknown locations based on known surrounding data [28]. In the context of this study, interpolation is specifically employed to alter the spatial resolution of images. The following sections describe the three most popular interpolation methods.

2.2.1.1 Nearest-neighbor Interpolation

Nearest-neighbor interpolation is the most basic and computationally efficient image interpolation technique. In this method, the value of a pixel in the resized image is assigned to be the value of the nearest pixel in the original image. The selection is based solely on the shortest Euclidean distance between the coordinates of the new pixel and those of the existing pixels [28].

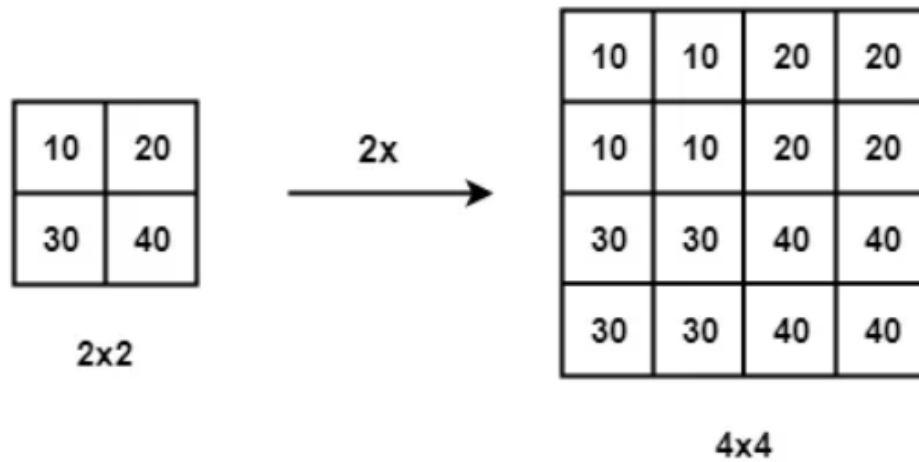


Figure 2.1: Example of a nearest neighbor interpolation with an upsampling factor of 2, applied on a 2x2 array. [5]

This approach does not consider any form of averaging or smoothing, making it extremely fast and suitable for applications where processing speed is more important than image quality. However, due to its discrete and abrupt pixel assignment, nearest-neighbor interpolation can result in visual artifacts such as jagged edges, blocky textures, and stair-step patterns, especially when enlarging images.

2.2.1.2 Bilinear Interpolation

Bilinear interpolation is a resampling technique used in image processing to estimate the intensity value of a pixel based on a weighted average of the four nearest pixels in the input image. Specifically, it considers the pixel values from a 2×2 neighborhood surrounding the target location and performs linear interpolation first in one direction, and then again in the other direction [28].

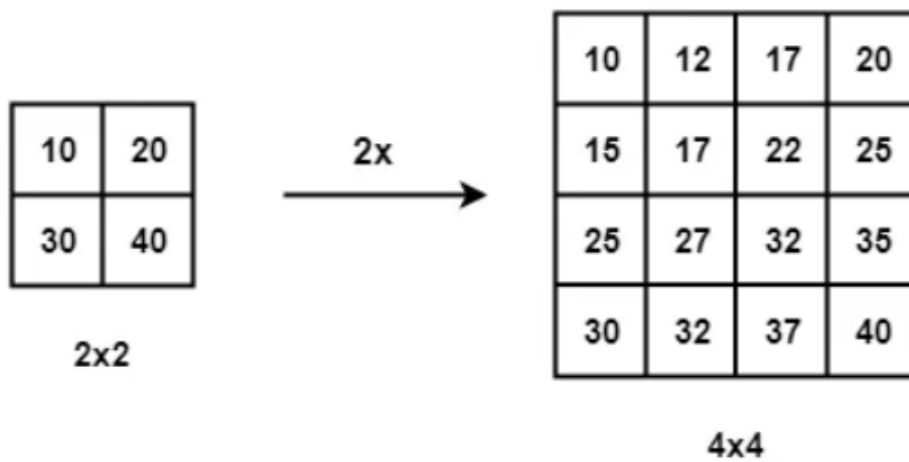


Figure 2.2: Example of a bilinear interpolation with an upsampling factor of 2, applied on a 2×2 array. [5]

This method provides a smoother and more visually continuous result than nearest-neighbor interpolation by factoring in both the spatial position and the values of adjacent pixels.

2.2.1.3 Bicubic Interpolation

Bicubic interpolation is a more advanced and accurate method of image resampling that extends the principles of bilinear interpolation by considering not just the closest 2×2 pixel neighborhood but a larger 4×4 grid (16 pixels) around the target coordinate. The method fits a smooth bicubic polynomial surface through these 16 points to estimate the value at a non-integer position in the output image [28].

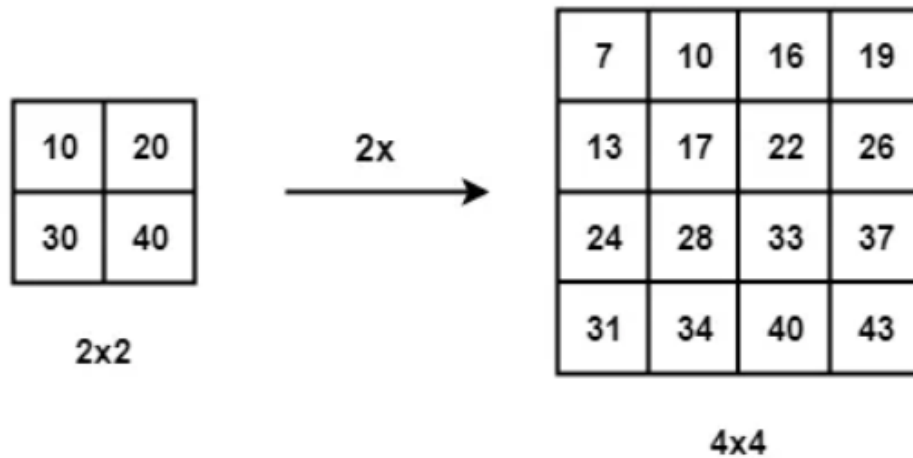


Figure 2.3: Example of a bicubic interpolation with an upsampling factor of 2, applied on a 2x2 array. [5]

Unlike nearest-neighbor or bilinear interpolation, bicubic interpolation accounts for both the pixel values and their gradients (or rate of change), which allows for a smoother and more continuous surface fit. This results in higher-quality images with reduced aliasing and fewer artifacts, especially when enlarging an image.

2.3 Evaluation Metrics

Evaluating the performance of SR algorithms is essential for assessing their effectiveness and guiding further improvements. Classical metrics such as Mean Squared Error (MSE), Mean Absolute Error (MAE), Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) are commonly employed due to their simplicity and computational efficiency. [29] These metrics provide a baseline for quantifying both numerical accuracy and perceptual quality. The following are the definitions of the metrics used in this study.

2.3.1 Mean Squared Error (MSE)

Mean Squared Error (MSE) is a straightforward metric that calculates the average squared difference between the pixel values of the estimated HR image and ground-truth image. It is defined as: [30]

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2.1)$$

where y_i and $\hat{y}_i$ are the pixel values of the two images, and n is the total number of pixels. MSE is computationally efficient and easy to implement, making it a popular choice for evaluating SR algorithms. However, it can be sensitive to small changes in pixel values that may not be perceptually relevant, leading to poor correlation with human perception. Lower MSE values indicate better image quality. [29]

2.3.2 Mean Absolute Error (MAE)

Mean absolute error (MAE) is defined as the absolute difference between intensities in pixels of the estimated HR image and ground-truth image. It is defined as: [30]

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2.2)$$

where y_i and $\hat{y}_i$ are the pixel values of the two images, and n is the total number of pixels.

The MAE is a popular metric because the error value units match the predicted target value units. In MAE, different errors are not weighted more or less, but the scores increase linearly with the increase in errors. In contrast, MSE penalizes larger errors more, inflating or increasing the mean error value as a result of the square of the error value. [31]

2.3.3 Peak Signal-to-Noise Ratio (PSNR)

The Peak Signal-to-Noise Ratio (PSNR) is a measure of fidelity that is used to evaluate the quality of reconstructed images by comparing the ground truth image's

peak signal value with the super-resolution algorithm's error. It is based on the MSE between the original and the reconstructed images. Mathematically, PSNR is defined as: [32]

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{L^2}{\text{MSE}} \right) \quad (2.3)$$

Where:

- L is the maximum possible pixel value (e.g., 255 for 8-bit images).
- MSE is the mean squared error between the original and the reconstructed image.

PSNR is commonly used to assess the performance of SR algorithms due to its computational efficiency and ease of implementation. However, it has been found to have limited relevance to human perception, as it does not take into account structural and perceptual dissimilarities between images. [29] PSNR is measured in decibels (dB), with higher values indicating better image quality.

2.3.4 Structural Similarity Index Measure (SSIM)

The Structural Similarity Index (SSIM) measures the similarity between two images by comparing their luminance, contrast, and structure. Typically, it compares two images - an estimated HR image and a ground truth HR image - by analyzing the mean, standard deviation, and cross-covariance of pixel intensities within local windows. SSIM between two image patches x and y is defined as: [33]

$$\text{SSIM}(x, y) = [l(x, y)]^\alpha \cdot [c(x, y)]^\beta \cdot [s(x, y)]^\gamma \quad (2.4)$$

The equation uses different variables to compare two images. l stands for lumi-

nance, which is used to compare the brightness of the images. c is contrast, which helps to distinguish between the brightest and darkest regions of the images. s is structure and it compares the local luminance patterns of the images to determine their similarity or dissimilarity. [29]

Typically, $\alpha = \beta = \gamma = 1$, so the formula simplifies to:

$$\text{SSIM}(x, y) = l(x, y) \cdot c(x, y) \cdot s(x, y) \quad (2.5)$$

The individual components are defined as:

$$l(x, y) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \quad (2.6)$$

$$c(x, y) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \quad (2.7)$$

$$s(x, y) = \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3} \quad (2.8)$$

Where:

- μ_x, μ_y are the local means of x and y
- σ_x, σ_y are the local standard deviations
- σ_{xy} is the local cross-covariance
- C_1, C_2 , and C_3 are small constants to stabilize the division, typically:

$$C_1 = (K_1L)^2, \quad C_2 = (K_2L)^2, \quad C_3 = \frac{C_2}{2}$$

SSIM is a more appropriate metric than PSNR for evaluating the perceived quality of SR algorithms, as it correlates better with human perception. However,

SSIM is computationally more demanding than PSNR, as it requires local statistics computation.[29] The closer the value is to 1, the closer the output image is to the original HR image.

Chapter 3

Background and Related Work

This chapter presents an overview of the theoretical background and related work that underpins this study. It briefly outlines key concepts in SR and climate downscaling, with a particular focus on recent advances in DL methods. The reviewed literature serves as a foundation for the methodological choices made in this work and highlights relevant developments that guide the proposed approach.

3.1 Single Image Super-Resolution Overview

As previously noted, this study addresses the challenge of climate data spatial downscaling through the lens of classical SISR. Classical SISR refers to the task of reconstructing high-resolution (HR) images from low-resolution (LR) inputs that have typically been downsampled using bicubic interpolation or subjected to unknown degradation processes. Standard magnification factors in such tasks include $\times 2$, $\times 3$, and $\times 4$.

A high-level categorization of classical SISR methods, based on broader conceptual families, is as follows [34]:

1. **Interpolation-Based Methods:** These include techniques such as bicubic interpolation [32] and Lanczos resampling [35]. While these methods are computationally efficient and simple to implement, they generally suffer from limited accuracy and poor reconstruction of fine details.

2. **Reconstruction-Based Methods:** Approaches such as Total Variation (TV) regularization [36] and Gradient Profile Prior [37] incorporate prior knowledge or constraints to restrict the solution space, enabling the generation of sharper and more flexible reconstructions. However, their performance often degrades at higher upscaling factors, and the associated computational cost can be substantial.
3. **Learning-Based Methods:** Also referred to as example-based methods, these utilize statistical relationships between LR and HR image pairs learned from large datasets. They typically offer a balance between accuracy and computational efficiency. Notable techniques include the Markov Random Field (MRF) model [38] and neighbor embedding methods [39].
4. Lastly, **Deep Learning-Based SISR** algorithms have demonstrated great superiority to reconstruction-based and other learning-based methods. SRCNN [7] was the first deep learning-based approach to image super-resolution and marked a significant breakthrough in the field. It outperformed traditional non-deep-learning methods in terms of Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and inference speed. The success of SRCNN demonstrated the strong potential of deep learning for SISR tasks and paved the way for further advancements. Since then, a wide range of models have been proposed, leading to substantial improvements in performance. Modern SISR approaches are primarily based on Convolutional Neural Networks (CNNs), Generative Adversarial Networks (GANs), and Transformer architectures, each offering distinct strategies for learning complex, nonlinear mappings from low to high resolution images.

The following provides a brief overview of these approaches along with their most representative models [2].

3.1.1 Convolutional Neural Network (CNN)-based

CNN consists of convolutional layers, activation functions, and pooling layers. Thanks to advancements in computer technology, CNN is now widely used in the field of SISR. The first algorithm to use CNN in the SISR domain was SRCNN [2].

Building on SRCNN, subsequent models introduced architectural and optimization improvements. FSRCNN [40] redesigned the network into an hourglass shape for speed and efficiency, using deconvolution layers and smaller kernels to achieve over $40\times$ faster performance than SRCNN. VDSR [41] further enhanced accuracy by significantly increasing network depth, though this came with higher training complexity due to the growing number of parameters.

To address these challenges, SRResNet [42] incorporated residual connections to ease training and reduce memory demands, while EDSR [8] refined this approach by removing Batch Normalization layers and introducing residual scaling, resulting in improved stability and performance.

Despite substantial gains in metrics like PSNR and SSIM, CNN-based SISR models still face issues such as over-smoothed outputs, limited receptive fields, and high computational costs due to their deep architectures [2].

3.1.2 Generative Adversarial Network (GAN)-based

CNN-based SISR models often produce overly smooth images, lacking high-frequency details. To address this, SRGAN [42] introduced Generative Adversarial Networks (GANs) into the field by using a generator to produce HR images from LR inputs and a discriminator to distinguish between real and generated images. SRGAN employs SRResNet as its generator and uses an adversarial loss in combination with a VGG-based perceptual loss, replacing the traditional MSE loss. While this leads to lower PSNR and SSIM, it significantly improves mean opinion score (MOS), resulting in images that appear more realistic to human observers.

However, SRGAN suffers from training instability and complexity. To overcome these limitations, ESRGAN [43] was proposed, introducing several architectural and training improvements. It removes Batch Normalization (BN) layers, incorporates Residual-in-Residual Dense Blocks (RRDBs), and adopts a relativistic discriminator to better judge image quality. Additionally, ESRGAN refines the perceptual loss by using VGG feature maps before activation, enhancing texture and detail reconstruction. These enhancements make ESRGAN more effective, stable, and visually impressive.

While GAN-based SISR models excel at generating perceptually realistic images, they still face challenges, including training instability, high computational demands, and difficulty in evaluating image quality objectively [2].

3.1.3 Transformer-based models

Transformers, originally introduced in 2017 for natural language processing, rely entirely on self-attention mechanisms, eliminating the need for recurrence and convolutions. Their ability to capture global dependencies has led to outstanding performance in sequence modeling, inspiring their application in computer vision—including Single Image Super-Resolution (SISR).

SwinIR [44] was the first notable Transformer-based model in SISR, built upon the Swin Transformer architecture. It combines the strengths of both CNNs and Transformers and is structured in three stages: shallow feature extraction, deep feature extraction (via Residual Swin Transformer Blocks), and image reconstruction. This hybrid architecture improves translational consistency and enables multi-level feature integration. SwinIR outperforms traditional CNN-based models in reconstruction quality but requires significant computational resources and GPU memory.

To improve the efficiency of Transformer-based SISR models many other lightweight and resource-friendly designs have been developed. These models aim to reduce memory consumption and computational cost while maintaining high reconstruction quality. By optimizing attention mechanisms and integrating efficient structural elements, newer approaches achieve a better balance between performance and resource usage, making Transformer-based SISR more practical for real-world applications [2].

Despite their excellent performance, Transformer-based SISR models remain resource-intensive, with high training costs, slower convergence, and greater architectural complexity compared to CNN-based approaches [2].

3.1.4 Performance Comparison

Table 3.1 presents a comparison of PSNR and SSIM results for various representative SISR models evaluated on widely used benchmark datasets: Set5, Set14, BSD100, and Urban100. Set5 is a small dataset with 5 high-quality images, while Set14 includes 14 images and is slightly more diverse. Both are commonly used to evaluate

3.1 : Single Image Super-Resolution Overview

super-resolution methods. BSD100, a subset of the larger Berkeley Segmentation Dataset, includes 100 test images frequently used for image restoration tasks. Urban100 focuses on urban scenes, such as buildings and street views, and is designed to challenge models with complex structures. These results are compiled from this study [2], offering a comprehensive performance comparison across different models.

Model	Scale	Set5	Set14	BSD100	Urban100
SRCNN	×2	33.66 / 0.9542	32.45 / 0.9067	31.36 / 0.8879	29.50 / 0.8946
	×3	32.75 / 0.9090	29.30 / 0.8215	28.41 / 0.7863	26.24 / 0.7989
	×4	30.09 / 0.8627	27.18 / 0.7861	26.26 / 0.7291	24.52 / 0.7221
FSRCNN	×2	37.05 / 0.9560	32.66 / 0.9090	31.53 / 0.8920	29.88 / 0.9020
	×3	33.18 / 0.9140	29.37 / 0.8240	28.53 / 0.7910	26.43 / 0.8080
	×4	30.71 / 0.8657	27.59 / 0.7535	26.98 / 0.7398	24.62 / 0.7280
VDSR	×2	37.53 / 0.9588	33.03 / 0.9124	31.90 / 0.8960	30.76 / 0.9140
	×3	33.68 / 0.9201	29.86 / 0.9312	28.83 / 0.7966	27.15 / 0.8315
	×4	31.35 / 0.8838	28.01 / 0.7674	27.29 / 0.7251	25.18 / 0.7524
SRResNet	×4	32.05 / 0.9019	28.49 / 0.8184	27.58 / 0.7620	26.07 / 0.7839
SRGAN	×4	29.40 / 0.8472	26.02 / 0.7397	25.16 / 0.6688	–
EDSR	×2	38.11 / 0.9602	33.92 / 0.9195	32.32 / 0.9013	32.93 / 0.9351
	×3	34.65 / 0.9280	30.52 / 0.8462	29.25 / 0.8093	28.80 / 0.8653
	×4	32.46 / 0.8968	28.80 / 0.7876	27.71 / 0.7420	26.64 / 0.8033
SRDenseNet	×4	32.02 / 0.8934	28.50 / 0.7782	27.53 / 0.7337	26.05 / 0.7819
RCAN	×2	38.27 / 0.9614	34.12 / 0.9216	32.41 / 0.9027	33.34 / 0.9384
	×3	34.74 / 0.9299	30.51 / 0.8461	29.32 / 0.8111	29.09 / 0.8702
	×4	32.63 / 0.9002	28.87 / 0.7889	27.80 / 0.7442	26.82 / 0.8087
ESRGAN	×4	32.73 / 0.9011	28.99 / 0.7917	27.85 / 0.7455	27.03 / 0.8153
HAN	×3	34.75 / 0.9299	30.67 / 0.8483	29.32 / 0.8110	29.10 / 0.8705
	×4	32.64 / 0.9002	28.90 / 0.7890	27.80 / 0.7442	26.85 / 0.8090
SwinIR	×2	38.14 / 0.9611	33.86 / 0.9206	32.31 / 0.9012	32.76 / 0.9340
	×3	34.62 / 0.9289	30.54 / 0.8463	29.20 / 0.8082	28.66 / 0.8624
	×4	32.44 / 0.8967	28.77 / 0.7858	27.69 / 0.7406	26.47 / 0.7990
ESRT	×4	34.42 / 0.9268	30.45 / 0.8433	29.15 / 0.8063	28.46 / 0.8574
EMT	×3	34.80 / 0.9303	30.71 / 0.8489	29.33 / 0.8113	29.16 / 0.8761
	×4	32.64 / 0.9003	28.97 / 0.7901	27.81 / 0.7441	26.98 / 0.8105

Table 3.1: PSNR/SSIM results of different SISR models on standard benchmark datasets [2].

3.2 Regional Climate Downscaling Approaches

To generate HR climate information from coarse-resolution GCMs, a variety of downscaling methods have been developed. These methods aim to bridge the spatial scale gap between global simulations and the localized climate information needed for impact assessments and decision-making. Typically, the resolution of GCMs ranges from 50 to 300 km, while RCMs aim to achieve spatial resolutions between 1 and 50 km. Broadly, regional climate downscaling approaches can be categorized into four main types:

1. **Dynamical Downscaling:** Traditional methods that use coarse-resolution global climate models (GCMs) as boundary conditions for high-resolution regional climate models (RCMs). These approaches rely on physical laws to simulate local climate processes.
2. **Statistical Downscaling:** Approaches that develop empirical relationships between observed historical climate data and model outputs (typically from GCMs) over the same time period. These relationships are then used to project future local climate conditions from large-scale model outputs.
3. **Statistical–Dynamical (Hybrid) Downscaling:** Methods that combine both statistical and dynamical techniques. This may involve using statistical models to post-process RCM outputs, or applying nudging techniques to align RCM dynamics with large-scale GCM fields.
4. **RCM Emulators:** Emerging data-driven approaches that leverage machine learning to emulate the behavior of RCMs. These models take coarse-resolution global inputs (e.g., from GCMs or reanalysis) and generate high-resolution regional outputs, mimicking RCM performance with significantly reduced computational cost. RCM emulators aim to integrate the strengths of both dynamical and statistical downscaling methods [45].

RCM emulators can be categorized into three main approaches based on the source of the predictors and the nature of the target data.

1. **Perfect Prognosis (PP)**: In this approach, the machine learning model uses coarse-resolution variables from a RCM (typically upscaled) as predictors and high-resolution outputs from the same RCM as targets [46]. Since both inputs and outputs originate from the same model, this setup maintains internal consistency and allows the emulator to learn physically meaningful relationships. Additionally, because PP emulators are trained on the RCM’s internal dynamics, they may generalize well to other GCMs, particularly when the RCM physics remains consistent across different GCM forcings. However, a limitation of this approach is that it does not account for GCM-specific biases or errors, as the emulator is not exposed to actual GCM outputs during training. If the physical processes or biases in the RCM diverge significantly from those in the driving GCM, the emulator’s predictions may be less accurate [47].

2. **Imperfect approach or Model Output Statistics (MOS)**: In this approach, the emulator uses large-scale predictors from the driving GCM (not from the RCM) and aims to reproduce high-resolution outputs from the RCM [46]. This setup introduces a mismatch between the input and target models, making the task more challenging but also more representative of real-world applications, especially for future climate projections. The term “imperfect” reflects this discrepancy, requiring the emulator to learn how to translate the GCM’s coarse outputs into the finer-scale RCM representation. One advantage of this method is its potential to correct GCM-specific biases, since it is explicitly trained on GCM inputs. However, a key limitation is that the emulator often becomes tied to the specific characteristics of the GCM, which can reduce its transferability to other GCMs [47].

3. **Hybrid Imperfect Approach (HIA)**: HIA combines elements of both PP and MOS, aiming to leverage the strengths of each approach while mitigating their weaknesses. This more flexible setup uses observed high-resolution climate data as targets (e.g., from satellite or in situ measurements), and coarse-scale variables from reanalysis data as inputs [48].

3.3 Review on Climate Downscaling Using DL

This section offers a brief overview of recent DL-based approaches for climate downscaling, particularly focusing on SR techniques aimed at enhancing the spatial resolution of climate data. While the field is extensive and rapidly evolving, the studies highlighted here represent a selection of recent contributions that have provided methodological insights and inspiration for the present work. These selected works showcase a range of modeling strategies and data integration techniques that aim to improve the accuracy, realism, and applicability of downscaled climate outputs.

Chiang et al. (2024) [49] propose a DL-based SR model for downscaling daily precipitation data over Taiwan from coarse-resolution ERA5 (25km) to fine-scale observational data (5km). The model architecture combines convolutional layers with skip connections and attention mechanisms to enhance spatial detail. Moreover, it integrates auxiliary data such as elevation—log-transformed to align with the skewed distribution of precipitation—enabling the model to better capture orographic effects and spatial variability. Their results demonstrate superior performance over baseline methods across multiple metrics, including MAE, RMSE, Pearson correlation, and SSIM, underscoring its effectiveness in hydrologically complex regions.

Expanding the application of generative models, Merizzi et al. (2024) [50] introduce a diffusion-based super-resolution framework for downscaling wind speed from ERA5 (25km) to CERRA (5km) over Italy. Their model employs a denoising U-Net within a diffusion process, trained without auxiliary static inputs. Despite this, it successfully reconstructs fine-scale wind patterns with greater physical consistency and spatial fidelity than traditional interpolation methods. Validation against in-situ station data reveals that the model achieves significantly higher SSIM scores and improved spatial realism, suggesting its utility as a data-filling tool for operational applications.

Building on this, Tomasi et al. (2025) [51] propose an enhanced diffusion-based approach that integrates HR static fields (digital elevation models (DEM), land cover types, and latitude) to guide the SR process. These geographic priors provide crucial spatial context, enabling the model to better represent complex terrain and

land-surface interactions. Their inclusion not only improves spatial accuracy and distributional fidelity but also enhances the model’s ability to generate physically consistent climate fields. This work highlights the critical role of incorporating static physical features in deep learning-based downscaling, especially when targeting fine-scale regional patterns.

Among recent studies, Jha et al. (2025) [52] develop and evaluate a deep learning-based approach for temperature downscaling using residual CNNs. The study explores the effectiveness of SR techniques in improving the spatial resolution of temperature data above Central Asia, the Indian subcontinent, and the Arabian Peninsula. The authors utilize ERA5 reanalysis data and create training pairs by artificially degrading the high-resolution data (originally at 0.25° resolution) via bi-linear interpolation to a coarse resolution of $2.5^\circ \times 2.5^\circ$. A key strength of the work lies in its extensive experimental framework, which compares various architectural configurations (SRCNN, VDSR, and EDSR) and training settings. The authors emphasize the importance of incorporating HR static auxiliary inputs, such as elevation, to better capture fine-scale temperature gradients, particularly in mountainous regions. Their findings demonstrate that residual learning significantly boosts model performance and spatial detail fidelity, supporting the utility of DL-based SR models for temperature downscaling in diverse terrain.

Together, these studies demonstrate the growing sophistication of DL-based downscaling methods (from convolutional architectures to diffusion models) and emphasize the importance of auxiliary geographic information for enhancing model performance and generalizability across diverse climate applications.

3.4 Summary and Connection to Methodology

Based on the general review of related work presented in this chapter, the methodological framework adopted in this study was carefully formulated and guided by empirical evidence from the recent literature.

To begin with, among the wide array of models proposed for SISR, CNN-based architectures were initially selected. This decision was primarily influenced by two

factors: the limited computational resources available and the demonstrated effectiveness of CNNs in achieving competitive results. As shown in Table 3.1, CNN-based models offer performance comparable to more complex approaches such as GANs and Transformer-based models, while maintaining lower computational overhead.

In terms of data formulation, the study adopts a Perfect Prognosis (PP) approach for defining predictors and targets. Specifically, HR outputs from the ERA5 reanalysis dataset are used as targets, while upscaled LR versions, created using bicubic interpolation, serve as predictors. This setup ensures internal consistency by deriving both input and output from the same source, eliminating inter-model discrepancies. Importantly, this design choice renders the method agnostic to any specific GCM, as it avoids the potential biases introduced by an Imperfect (MOS) approach that would require reliance on GCM-specific outputs.

Additionally, the review of recent DL applications in climate data downscaling revealed the effectiveness of architectural enhancements such as residual connections and attention mechanisms. These insights led to the evaluation of advanced CNN variants such as EDSR and RCAN, which incorporate these features to improve the learning of fine-grained spatial patterns. The literature also informed the selection of widely accepted evaluation metrics, with MAE, MSE, PSNR, and SSIM consistently used across comparable studies to assess model precision and visual fidelity.

Finally, the review highlighted the beneficial role of static auxiliary data, such as topographic and land use information, in improving the accuracy of downscaling. This insight has influenced the consideration of how land morphology and related geographic features can be effectively integrated into the modeling process to improve spatial realism and physical consistency.

Chapter 4

Deep Learning DL Models

This Chapter introduces and defines the deep learning model architectures employed in this study. It provides a brief overview of the underlying principles and design logic behind each model. Detailed architectural configurations and implementation strategies will be presented in the subsequent *Methodology* section.

4.0.1 Convolutional Auto-Encoder (CAE)

An autoencoder is a type of neural network specifically designed to compress input data into a lower-dimensional, meaningful representation and then reconstruct it as closely as possible to the original input. Initially introduced in [53], autoencoders are trained to replicate their input, enabling them to learn data representations in an unsupervised manner. These learned representations, often referred to as embeddings or latent features, can be useful for a variety of downstream tasks such as clustering. The figure 4.1 illustrates the general architecture of an autoencoder model. [6]

A Convolutional Autoencoder (CAE) is a specialized variant of the autoencoder architecture that employs convolutional layers in both its encoder and decoder components. While traditional autoencoders may incorporate fully connected layers, CAEs are particularly well-suited for image data due to their ability to capture spatial hierarchies and local features. Using the strengths of convolutional neural networks, CAEs can effectively exploit the inherent structure of images. [54] This

makes them especially suitable for image processing tasks, which is why they were selected for use in this SISR study.

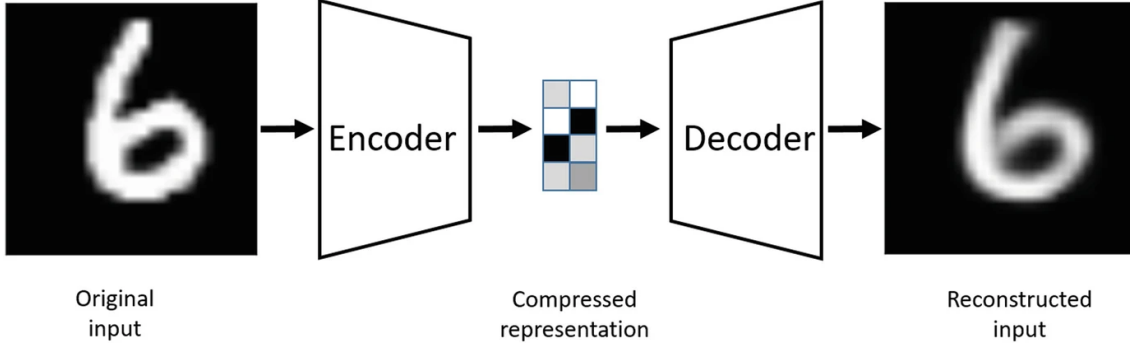


Figure 4.1: An autoencoder example. The input image is encoded to a compressed representation and then decoded. [6]

4.0.2 Super-Resolution Convolutional Neural Network (SRCNN)

Super-Resolution Convolutional Neural Networks (SRCNN) are deep learning models designed to enhance image resolution by learning a non-linear mapping from low- to high-resolution images. [7, 55] These networks outperform traditional up-sampling methods like bicubic interpolation in both speed and quality. SRCNN architectures typically involve multiple convolutional layers, each responsible for a specific functional stage in the super-resolution process.

1. The first layer performs patch extraction and representation, where overlapping image patches are extracted from the low-resolution input and mapped into a high-dimensional feature space to capture complex structures.

$$F_1(Y) = \max(0, W_1 * Y + B_1) \quad (4.1)$$

where Y is the input low-resolution image (typically upsampled via bicubic interpolation), W_1 and B_1 are the filter weights and biases of the first layer, and $*$ denotes the convolution operation. The ReLU function is used as the activation.

2. The second layer carries out non-linear mapping, transforming the feature representations of low-resolution patches into high-resolution counterparts.

$$F_2(Y) = \max(0, W_2 * F_1(Y) + B_2) \quad (4.2)$$

where W_2 and B_2 are the weights and biases of the second convolutional layer, which maps the high-dimensional feature representation to another set of high-dimensional features.

3. Finally, the third layer handles reconstruction, where the high-resolution patches are aggregated and merged to form the final high-resolution image.

$$F(Y) = W_3 * F_2(Y) + B_3 \quad (4.3)$$

where W_3 and B_3 are the weights and biases of the final convolutional layer, and $F(Y)$ is the reconstructed high-resolution image output.

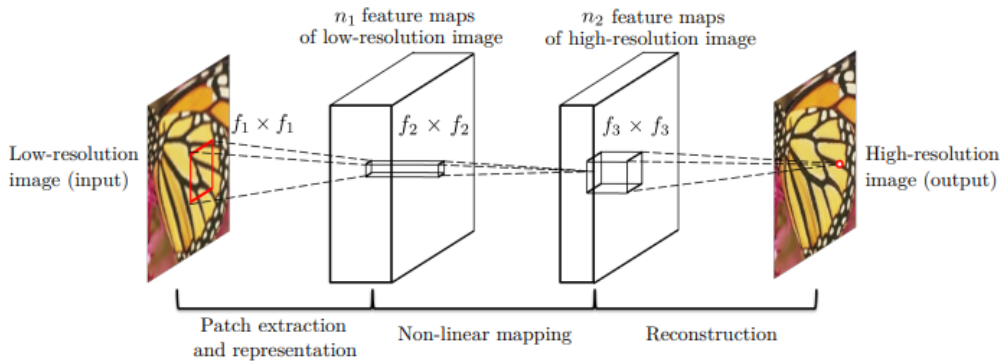


Figure 4.2: SRCNN Architecture. [7]

This end-to-end trainable architecture allows SRCNN to learn the optimal mapping directly from data, making it both effective and computationally efficient for image super-resolution tasks. An overview of the network is depicted in Figure 4.2. Given a low-resolution image Y , the first convolutional layer of the SRCNN extracts a set of feature maps. The second layer maps these feature maps nonlinearly to high-

resolution patch representations. The last layer combines the predictions within a spatial neighbourhood to produce the final high-resolution image $F(Y)$. [7]

4.0.3 Enhanced Deep Super-Resolution Network (EDSR)

The Enhanced Deep Super-Resolution (EDSR) network is a state-of-the-art convolutional neural network for improving image spatial resolution. [56] It builds upon residual learning by removing unnecessary components, such as batch normalization layers, which were shown to hinder performance in image restoration tasks. EDSR utilizes a very deep residual network with a simplified structure and greater model capacity, allowing it to learn more powerful mappings from low-resolution to high-resolution images. [8]

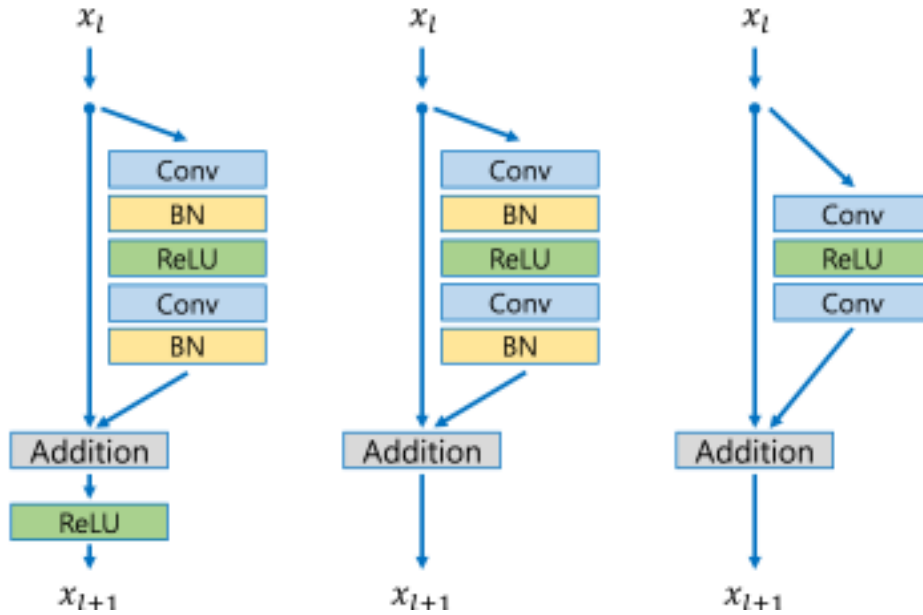


Figure 4.3: Comparison of residual blocks in original ResNet, SRResNet, and EDSR. [8]

Figure 4.3 shows the evolution from ResNet[57] to SRResNet[42] and eventually to EDSR[8] which reflects a series of architectural refinements aimed at optimizing residual networks for SISR tasks. While ResNet was originally designed for image classification, featuring residual blocks with batch normalization to ease the training of deep networks, its structure was not tailored for pixel-wise accuracy required in image restoration. SRResNet adapted the ResNet framework for super-resolution

by reconfiguring the output layers and retaining residual learning; however, it still included batch normalization layers, which limit performance by discarding essential image intensity information. To address these issues, the EDSR removed all batch normalization layers, increased the number of filters and residual blocks, and introduced residual scaling to stabilize training in deeper architectures [8]. These changes significantly improved the network's capacity to learn fine image details, making EDSR one of the most effective models for high-quality SISR. The architecture of the EDSR model is shown in Figure 4.4.

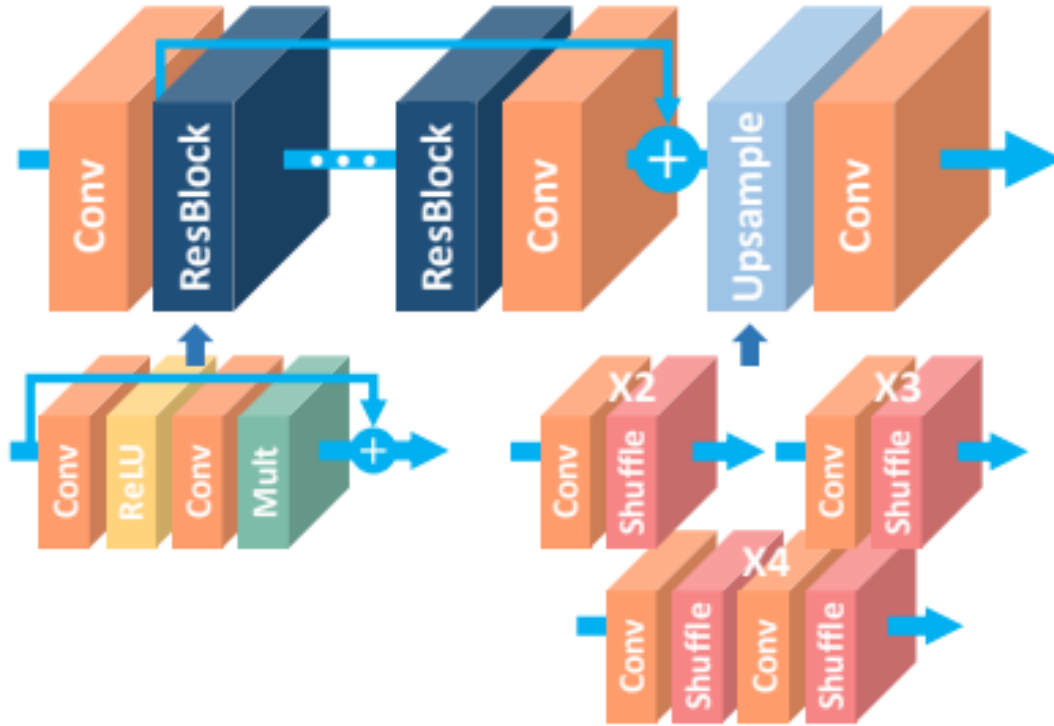


Figure 4.4: EDSR Architecture. [8]

4.0.4 Residual Channel Attention Network (RCAN)

The Residual Channel Attention Network (RCAN) is a powerful DL architecture, which is also specifically designed for SISR tasks. It extends the traditional residual network framework by incorporating two major innovations: deep residual-in-residual learning and a channel attention mechanism. The residual-in-residual structure allows the network to be very deep while maintaining training stability through

long skip connections across residual groups and blocks. Meanwhile, the channel attention module adaptively rescales channel-wise features by modeling interdependencies between feature maps, allowing the network to emphasize more informative channels and suppress less relevant ones. This selective attention is especially beneficial for restoring fine image details, such as textures and edges.

The overall architecture of the RCAN, including the residual-in-residual structure and the internal channel attention modules, is illustrated in Figure 4.5 and Figure 4.6. [9]

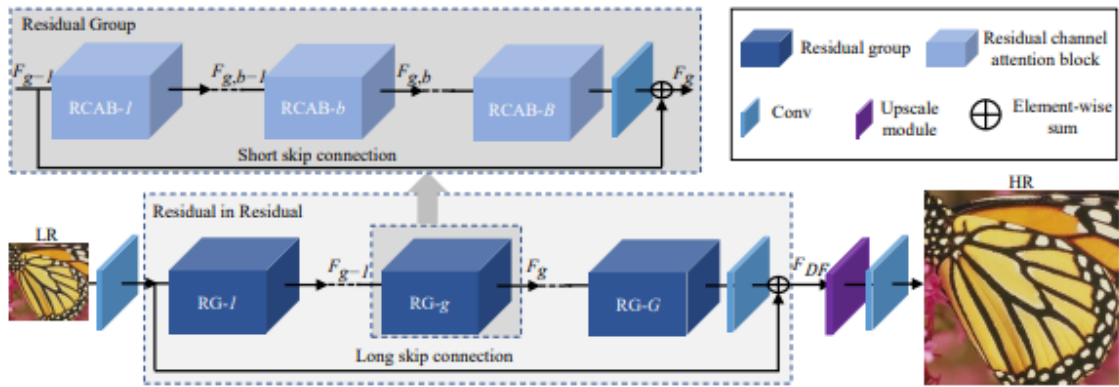


Figure 4.5: RCAN Architecture. [9]

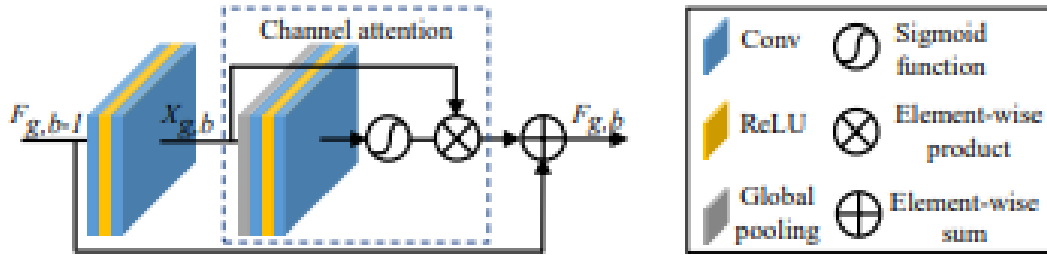


Figure 4.6: Residual channel attention block. [9]

Chapter 5

Methodology

This chapter outlines the methodology developed for this study, detailing the steps taken and the design decisions made. Each decision is justified to structure the proposed method to improve the spatial resolution of a climate data grid.

5.1 Data Collection and Preprocessing

The ground truth data used for training the DL models comes from the ERA5 dataset, available in the Copernicus Climate Data Store [3]. As has already been mentioned, ERA5 is the fifth-generation reanalysis by the ECMWF, providing global climate and weather data from 1940 to the present at hourly resolution.

This study focuses on the 2-meter air temperature (T2m), measured in kelvin (K), which represents air temperature 2 meters above land, sea, or inland waters. T2m was chosen because it is a key climate change indicator and influences various sectors of human activity. Compared to other parameters, such as wind, it has lower spatial variability, making it ideal for this study's first phase. The Table 5.1 provides a summary of the main characteristics of the dataset.

Data description

Data type	Gridded
Projection	Regular latitude-longitude grid
Horizontal coverage	Global
Horizontal resolution	Reanalysis: $0.25^\circ \times 0.25^\circ$ (atmosphere), $0.5^\circ \times 0.5^\circ$ (ocean waves)
Temporal coverage	1940 to present
Temporal resolution	Hourly
File format	GRIB
Update frequency	Daily

Table 5.1: ERA5 reanalysis Data description [3].

The pre-processing began by downloading GRIB files from the Copernicus Climate Data Store [3] for the years 2000 to 2020, with a 6-hour temporal resolution (00:00, 06:00, 12:00, and 18:00). The selected spatial grid covered latitudes 80.0° to 0.0° and longitudes -60.0° to 85.0° . These GRIB files were then converted into NPY files, organizing the data as time series to facilitate analysis that follows. Furthermore, the original spatial resolution of $0.25^\circ \times 0.25^\circ$ was upscaled to $0.5^\circ \times 0.5^\circ$ and $1^\circ \times 1^\circ$ using bicubic interpolation. As a result, three datasets were generated, each with different resolution and size, as shown in the Table 5.2.

Dataset	Resolution	Grid Size
High Resolution (HR)	$0.25^\circ \times 0.25^\circ$	320 x 580
Medium Resolution (MR)	$0.5^\circ \times 0.5^\circ$	160 x 290
Low Resolution (LR)	$1^\circ \times 1^\circ$	80 x 145

Table 5.2: Different Resolution Datasets Information.

Figure 5.1 presents the same random instance in the analysis of each of the three datasets mentioned above. In Figure 5.2, the same instance is zoomed in on the broader European region, making the differences in the analysis more apparent.

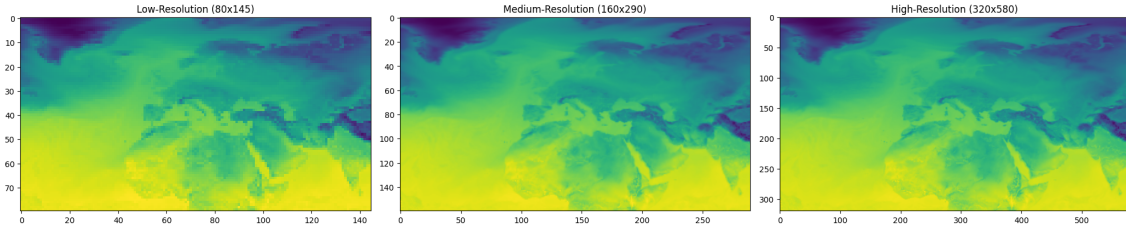


Figure 5.1: Visualization of the data for each resolution.

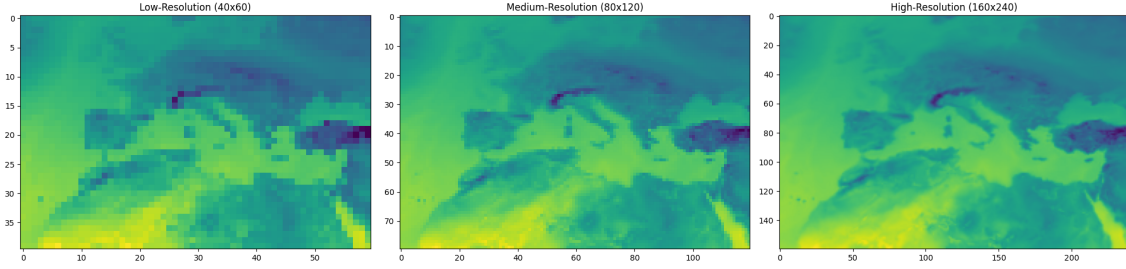


Figure 5.2: Visualization of the data for each resolution zoomed in Europe.

From the ERA5 dataset, the Land-Sea Mask parameter was also obtained. This parameter represents the proportion of land, versus ocean or inland waters (such as lakes, reservoirs, rivers, and coastal waters), within a grid cell, with values ranging from 0 to 1 (it is a dimensionless variable) [3]. As with T2m data, bicubic interpolation was applied to the Land-Sea Mask and the corresponding grids were produced to facilitate metric calculations only for the land portion of each grid cells. This focus on land is important because the temperature tends to fluctuate more there, making downscaling more challenging in land areas. Thus, by classifying any cell with a value greater than 0.5 as land, we obtain the three corresponding T2m datasets land mask grids shown in the Figure 5.3.

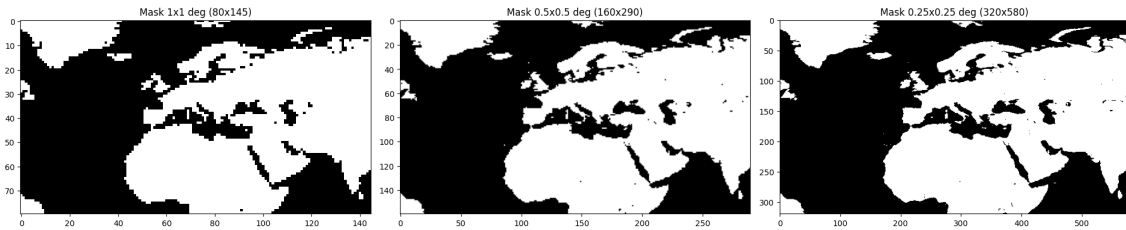


Figure 5.3: Land Mask Grids.

Altitude was selected as an additional feature to enhance the downscaling of T2m, as well as in areas with complex terrains elevation and topography signifi-

cantly influence temperature variations [58]. The altitude data was sourced from a Digital Elevation Model (DEM). DEM data was processed similarly to obtain elevation values for each grid cell. Using bicubic interpolation, the data were converted into three grids with resolutions of $0.25^\circ \times 0.25^\circ$, $0.5^\circ \times 0.5^\circ$, and $1^\circ \times 1^\circ$, respectively. The processed grids were then saved as NPY files for use in the following analysis. Figure 5.4 displays a visualization of these data, while Figure 5.5 offers a zoomed view of the broader European region to better highlight the differences in the analysis.

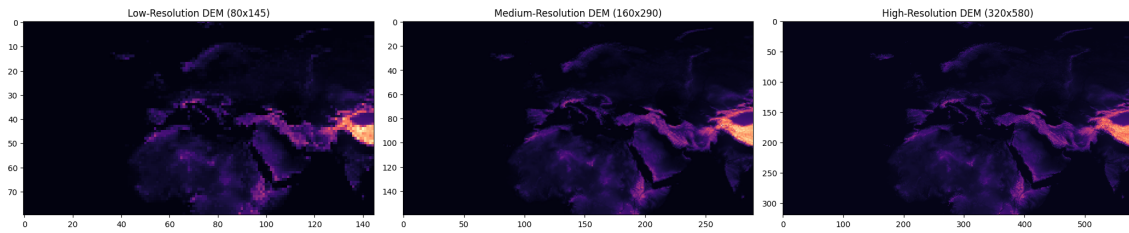


Figure 5.4: Visualization of the elevation data for each resolution.

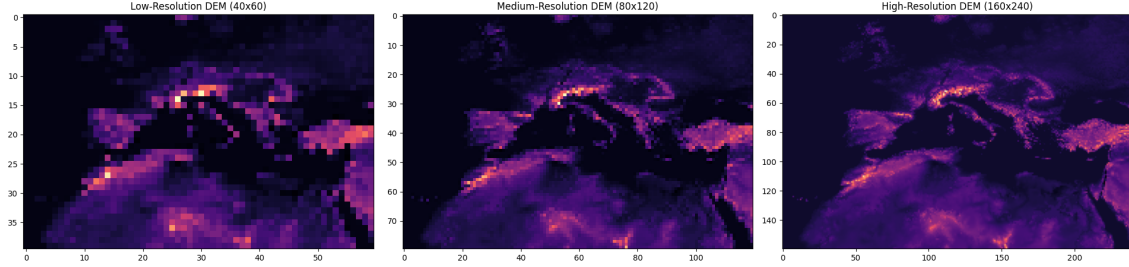


Figure 5.5: Visualization of the elevation data for each resolution zoomed in Europe.

Notably, the elevation data for the Greenland region included in the T2m data grid is missing because it was not available in the DEM model we used and proved difficult to integrate from alternative sources.

5.2 Data Preparation

Before testing any of the methods presented below, the data was pre-processed uniformly to ensure fair comparison of results. Initially, the data was normalized and then split into three subsets: training, validation, and testing. The preprocessing methods used are detailed in the following subsections.

5.2.1 Normalization

For data normalization, the z-score technique was used to standardize the dataset, ensuring a normal distribution with values ranging between -1 and 1. This method required calculating the global mean and standard deviation of the data set. The obtained values are:

$$Mean = 286.30798 \text{ and } Std = 64.25680$$

Therefore, in each grid of the dataset before being used by any method this formula was applied:

$$z = (X - Mean)/Std$$

5.2.2 Dataset Splitting

To ensure a fair and reliable evaluation of the methods, the dataset was split into three subsets: 70% for training, 15% for validation, and 15% for testing. The training set is used to fit the models, the validation set helps tune hyperparameters and prevent overfitting, and the test set provides an unbiased final evaluation. To eliminate any temporal dependencies and ensure randomness, the data was shuffled before splitting, disrupting its original timeline. Additionally, a fixed random seed was used to make the process reproducible, allowing consistent partitioning across different experiments and ensuring fair comparison between models.

5.3 Baseline Methods

As a baseline and to assess whether the application of DL techniques for downscaling climate data has a significant impact on results, three simple downscaling methods were first applied: bicubic interpolation, nearest-neighbor interpolation, and a bias-corrected variant of nearest neighbor interpolation. The purpose of this comparison is to determine whether DL-based approaches provide meaningful improvements over traditional interpolation methods.

In the bias-corrected nearest neighbor method, an additional correction grid was introduced to refine the interpolated values. This correction grid was computed using the training dataset by first applying nearest neighbor interpolation to down-scale grids from $1^\circ \times 1^\circ$ to $0.25^\circ \times 0.25^\circ$ resolution. The differences between these interpolated grids and the ground truth were then averaged per cell to create a bias correction grid, which was subsequently added to the nearest neighbor interpolated results during testing.

All three methods were applied to the test dataset, and the evaluation metrics outlined in the evaluation strategy were calculated to ensure direct comparison with the DL-based models discussed in the following sections. The results of these methods are presented in Table 6.1, Chapter 5 (Results and Discussion). After analyzing the performance of the three approaches, the method that achieved the best results was selected as the benchmark for further comparison against the DL models.

5.4 Training and Validation Strategy

To ensure robust model performance a systematic training and validation strategy was employed. The following outlines the procedures followed for training configuration and evaluation during model development. The goal was to optimize performance while minimizing overfitting and ensuring fair comparison between model variants.

5.4.1 Validation Metrics and Loss Function

To validate the performance of the proposed methods for temperature grid down-scaling, different statistical metrics are used and specifically, MAE, MSE, PSNR and SSIM. Since this problem can be framed as a SISR task, these metrics effectively measure both pixel-wise accuracy and structural fidelity. MSE, the simplest and most widely used full-reference quality metric, quantifies the average squared differences between predicted and actual temperature values. PSNR, which is directly derived from MSE, assesses reconstruction quality by measuring the ratio between the maximum possible signal and the error level. These two metrics are popular

due to their ease of computation, clear mathematical interpretation, and usefulness in optimization. However, they do not always align with perceptual quality, which is why SSIM is included—it evaluates structural similarities by considering luminance, contrast, and spatial patterns [33]. Finally, MAE provides an intuitive measure of absolute errors, making it useful for interpreting real-world deviations in temperature values. Together, these metrics offer a comprehensive assessment of both numerical accuracy and structural preservation. The values were computed over the course of the entire training process, and their corresponding curves were plotted from the recorded model history upon completion.

MAE was chosen as the loss function for all DL models that there used instead of MSE because it provides a more robust and domain-specific measure of error for temperature data. MAE, also known as L1 loss, calculates the average of the absolute differences between predicted and actual values [59]. Unlike MSE, which squares these differences and amplifies the impact of large errors, L1 loss treats all errors linearly, making it less sensitive to outliers [60]. This is especially important in temperature grid downscaling, where extreme values or localized noise could distort the learning process if MSE were used. Additionally, MAE aligns better with real-world temperature assessment, as it directly represents the average deviation in the same units as the data [61]. This makes model performance easier to interpret and ensures that the focus remains on minimizing overall error across all regions rather than disproportionately penalizing large deviations, as MSE does.

5.4.2 Optimizer and Learning Rate Schedule

A DL optimizer is an algorithm used to adjust the weights and biases of a Neural Network (NN) to minimize the loss function during training. The goal is to find the set of parameters that allows the model to make the most accurate predictions. Optimizers do this by using gradient descent and its variants, which compute how the model's error changes with respect to each parameter and update the parameters in the opposite direction of the gradient. Well-known optimizers in deep learning encompass Stochastic Gradient Descent (SGD), Adam, and RMSprop, each equipped with distinct update rules, learning rates, and momentum strategies.

During this study all DL models that used were optimized using the Adam optimizer. Adam optimizer, short for Adaptive Moment Estimation optimizer, serves as an optimization algorithm commonly used in DL. It extends the SGD algorithm and updates the weights of a NN during training. Unlike SGD, which maintains a single learning rate throughout training, Adam optimizer dynamically computes individual learning rates based on the past gradients and their second moments. Adam is computationally efficient, has little memory requirement, invariant to diagonal rescaling of gradients, and is well suited for problems that are large in terms of data/parameters [62].

Even though Adam dynamically adapts the learning rate during the training process, training can sometimes stall or converge slowly. In such cases, manually reducing the learning rate can help fine-tune the model further, especially near local minima. Therefore, a ReduceLROnPlateau scheduler was applied to decrease the learning rate when the loss function stops improving for a certain number of epochs.

5.4.3 Batch Size and Epochs

A batch size of 128 was selected for model training based on initial testing. However, for cases involving downscaling from 0.5° to 0.25° , the batch size was reduced to 64, and further to 16 for methods that also incorporated DEM data, due to the increased dataset size and GPU RAM limitations. In all experiments, models were trained for 1000 epochs, with the exception of the initial architecture selection phase, where training was limited to 200 epochs to reduce computational costs while ensuring reliable conclusions.

Finally, the early stopping technique with restoration of the best weights was employed, triggered after 150 epochs without improvement in model performance. The only exception was again during the architecture selection phase, where the patience parameter was set to 50 epochs to reduce training time.

5.5 Implementation Details

All data preprocessing and DL model training were conducted using Google Colab notebooks, utilizing the A100 GPU with the high-RAM runtime option enabled. The dataset sizes varied significantly across resolution levels, with the LR dataset occupying approximately 2.7 GB, the MR dataset 10.9 GB, and the HR dataset reaching 21.2 GB. These substantial data volumes, especially at HR, imposed considerable demands on both memory and computational resources during training.

The implementation of the models was carried out using the TensorFlow [63] framework, which provided an efficient and flexible environment for building and training DL architectures. For data pre-processing tasks and data splitting, the scikit-learn (sklearn) library [64] was employed. Visualization of training progress, model performance metrics, and output comparisons was performed using Matplotlib [65], which enabled the generation of informative plots and grid visualizations.

Finally, code for tasks that were repeatedly used across different stages of the workflow, such as model training, visualization of evaluation metrics, and related operations, was modularized into reusable functions to enhance clarity and maintainability.

5.6 Post-Processing and Model Evaluation

To effectively evaluate each method, Land MAE was calculated in both normalized units and degrees Celsius to provide a meaningful interpretation of the temperature differences. Since temperature variations are strongly influenced by terrain and can fluctuate significantly throughout the day, MAE was specifically assessed over land areas to serve as a more objective evaluation criterion. This approach was chosen because large bodies of water, due to their high heat capacity, act as thermal buffers [66], reducing temperature fluctuations in nearby regions. As a result, the overall MAE can be disproportionately affected by ocean-covered areas, where temperature changes are much smaller compared to land. By focusing on land-only MAE, the evaluation better captures the model's ability to reconstruct temperature

variations in regions where changes are more pronounced and climatically relevant.

5.7 DL Model Architecture and Selection

To determine the most effective architecture for this study, four different DL models commonly used for SISR were tested. The architecture of each model, along with a comparison of their results, is presented in the following subsections. The library PlotNeuralNet [67] was used to draw the four NN architectures presented above.

5.7.1 CAE Architecture

The CAE model follows an encoder-decoder architecture designed for SISR. It takes a single-channel image as input and outputs a higher-resolution (double in this case), single-channel image.

- The **encoder** extracts hierarchical features from the input image using three convolutional layers with increasing filter sizes (64, 128, 256). Each layer applies batch normalization and a LeakyReLU activation, progressively down-sampling the spatial dimensions.
- A **bottleneck** layer flattens the feature maps and passes them through a dense layer (1024 neurons) to compress the data into a lower-dimensional representation.
- The **decoder** reconstructs the high-resolution image using transposed convolution layers (deconvolutions) that upsample the features back to the original size. Each upsampling step reduces the number of filters while maintaining structure with batch normalization and LeakyReLU activations.
- Finally, a **cropping** operation ensures the output size aligns with the desired resolution.

All convolutional and deconvolutional layers used 3×3 kernels. A detailed illustration of the architecture used to increase the resolution from 1° to 0.5° is shown

in Figure 5.6.

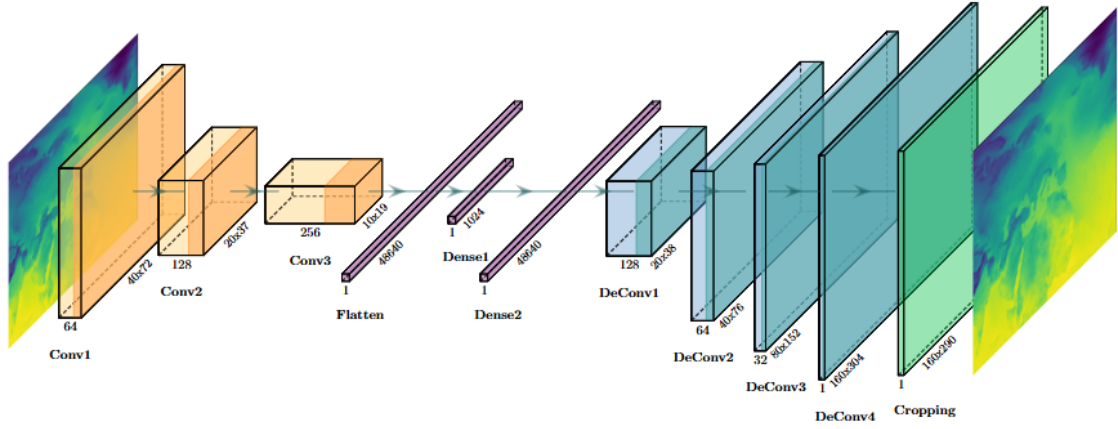


Figure 5.6: The architecture of the CAE implemented

5.7.2 SRCNN Architecture

The SRCNN is a simple and efficient approach for super-resolution tasks. The model selected for this study consists of three main layers [68]:

1. A **feature extraction layer**, which applies a 64-filter convolution with a large 9×9 kernel to capture low-level features from the input image.
2. A **non-linear mapping layer**, which refines the extracted features using a 5×5 convolution with 32 filters and a ReLU activation function.
3. And a **reconstruction layer**, which performs another 5×5 convolution to generate the final high-resolution output while maintaining the same spatial dimensions as the input.

A key difference between this architecture and the other three tested is that it requires an additional pre-processing step, as the input image must be resized to match the output size. The bicubic interpolation method was also used for this resizing and the implemented architecture is shown in Figure 5.7.

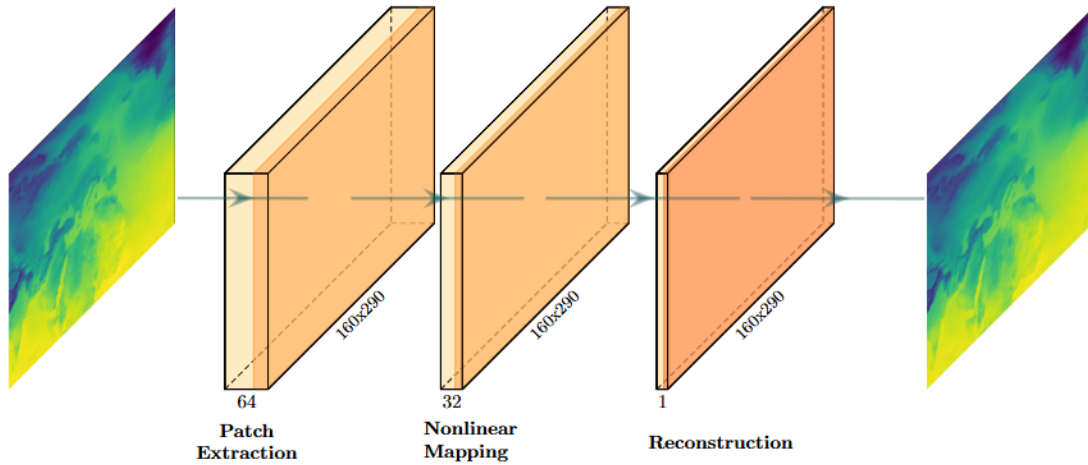


Figure 5.7: The architecture of the SRCNN implemented

5.7.3 EDSR Architecture

The EDSR model is built around residual learning, where multiple residual blocks extract features while preserving important image details. Each of the proposed model four residual blocks, consists of two 3×3 convolutional layers with a LeakyReLU activation and a scaling factor (0.1) to stabilize training. The model begins with an **initial convolutional layer** and passes through **four residual blocks**, followed by a final **skip connection** that adds the original features back to the processed ones. After feature extraction, **an upsampling layer doubles the resolution** of the input image. This architecture presented in Figure 5.8.

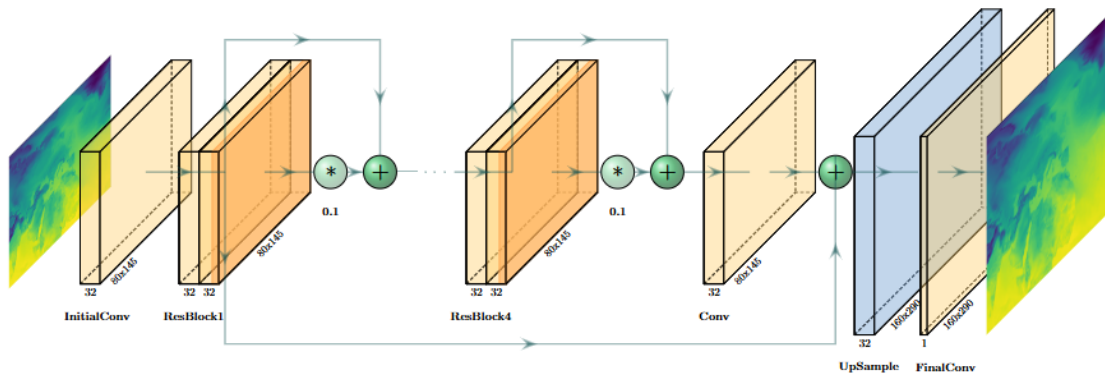


Figure 5.8: The architecture of the EDSR implemented

5.7.4 RCAN Architecture

The RCAN incorporates both residual learning and channel attention mechanisms. It consists of multiple residual blocks (four in this case), each enhanced with a channel attention module. The channel attention mechanism works by applying global average pooling to extract channel-wise statistics, followed by two fully connected layers that compute attention weights, which are then multiplied with the feature maps to emphasize important information. The model starts with an **initial convolutional layer**, followed by **four residual blocks with channel attention**, ensuring that critical features are preserved and enhanced. A **skip connection** adds the original features back to the processed ones before **upsampling**, which doubles the resolution of the input image. The architecture of the RCAN model implemented shown in Figure 5.9.

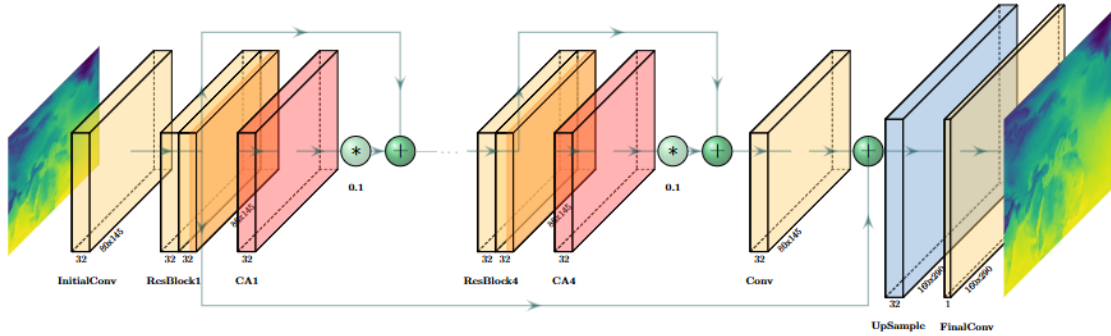


Figure 5.9: The architecture of the RCAN implemented

5.7.5 Architecture Selection

To select the best of the four architectures, each model was trained for 200 epochs to enhance the resolution of the input grid from $1^\circ \times 1^\circ$ to $0.5^\circ \times 0.5^\circ$. Early stopping was applied with a patience of 50 epochs and the learning rate was reduced on plateau after 40 epochs. The initial value of learning rate was 0.001 and the minimum value it is allowed to reach is 0.00001.

The values of the evaluation metrics for each architecture and the comparison between them are presented in Table 6.2 of Chapter 6 (Results and Discussion). The architecture that achieves the best overall performance is selected as the final

model to be used in all subsequent analyses.

5.8 Temporal and Resolution Strategy

To explore opportunities for improving model performance, two alternative strategies were evaluated to determine whether their inclusion in the final methodology would be beneficial.

The first approach investigated whether incorporating seasonal information could enhance predictive accuracy. Based on the findings of Karozis *et al* [11] suggesting that their CAE model benefited from seasonal distinctions, separate models were trained for each season and compared against a universal model trained on the entire dataset. All models were trained using the same training data (split by season in the case of the seasonal models) for 1000 epochs with a batch size of 128. Early stopping was applied with a patience of 150 epochs, and the learning rate was reduced after 100 epochs without improvement. The outcomes of this comparison are presented in Subsection 6.3 of Chapter 6 (Results and Discussion).

The second approach explored whether a two-step downscaling process, first from $1^\circ \times 1^\circ$ to $0.5^\circ \times 0.5^\circ$, and then from $0.5^\circ \times 0.5^\circ$ to $0.25^\circ \times 0.25^\circ$, could yield better results than a single-step model that performs the entire downscaling from $1^\circ \times 1^\circ$ to $0.25^\circ \times 0.25^\circ$ in one pass. This strategy was motivated by two main considerations:

1. Decomposing the task into simpler subtasks may reduce the learning burden for each model, making training more efficient and stable.
2. Each spatial scale may involve different dominant physical processes. For example, large-scale climate patterns prevail at coarser resolutions, while fine-scale phenomena such as orographic effects and land–atmosphere interactions become more prominent at higher resolutions [69]. Using two separate models allows each to better capture the distinct dynamics at their respective levels.

Models in this experiment were trained for 1000 epochs with a reduced batch size of 64, using early stopping (patience 150) and learning rate scheduling (ReduceLROnPlateau with patience 100). The results of this experiment are presented in Subsection 6.4 of Chapter 6.

5.9 Integration of Elevation Information

Informed by the literature review presented in Chapter 3, an additional method was explored to further enhance the performance of DL models in climate data downscaling. Specifically, related studies suggest that incorporating elevation information from Digital Elevation Models (DEMs) can positively influence prediction accuracy. To evaluate this, two different integration strategies were tested:

1. The first approach involved adding elevation data as a second input channel alongside the primary input, allowing the model to process spatial and elevation features in parallel. However, this method required the DEM data to be upscaled to match the resolution of the input grid, which may result in a loss of fine-grained elevation details.
2. To address this limitation, a second approach was implemented. Here, the high-resolution DEM data was introduced after the upsampling stage of the model, as a second channel to the intermediate output. This was followed by two fusion convolutional layers, which incorporated the elevation information into the final prediction. This method aimed to preserve the spatial richness of the DEM and allow for more meaningful integration into the model's output.

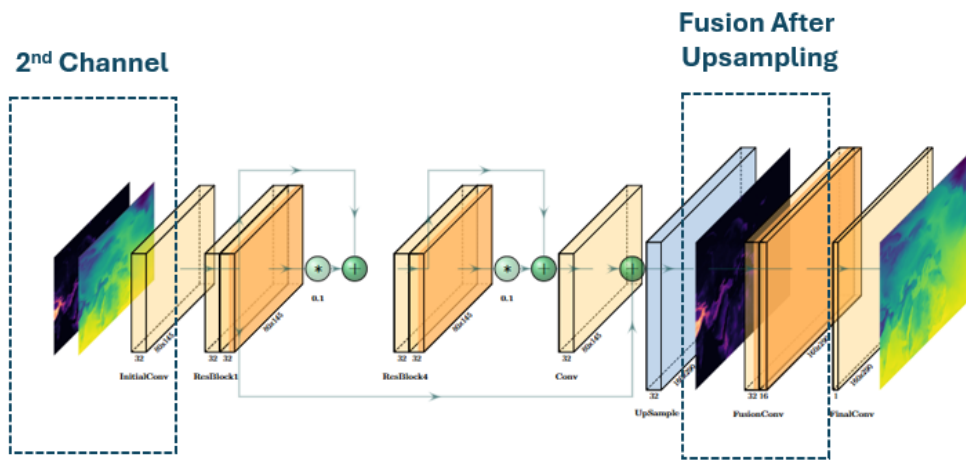


Figure 5.10: The architecture of the EDSR model with both DEM integration methods.

Finally, since each of the above strategies has its own limitations, a third model was developed that combines both approaches simultaneously. The architecture of this hybrid model is illustrated in Figure 5.10. This integrated design yielded the most significant improvement in performance, as demonstrated in Section 6.5 of the Results Chapter.

5.10 Summary and Transition to Results

In summary, the methodology followed in this study can be outlined in the following key steps. After preparing the dataset and selecting the evaluation metrics to be used consistently across all experiments, an initial comparison of various interpolation methods was performed to establish a baseline for evaluating the performance of the DL models. Next, a comparison of different architectures was conducted to select the most suitable one, followed by an assessment of whether separate seasonal models could yield better performance. Once the seasonal approach was deemed unnecessary, the potential benefits of performing downscaling in multiple stages were investigated. Finally, the integration of elevation data was explored to further enhance model accuracy.

The detailed results of these experiments and how they informed the step-by-step design decisions are presented in Chapter 6: Results and Discussion.

Chapter 6

Results and Discussion

In this chapter, the results obtained from each step of the study are presented and analyzed. The performance of different methods is evaluated using the selected metrics, allowing for a comparison between traditional and DL-based downscaling approaches. Additionally, the key decisions made at each stage of the process are justified based on the observed outcomes.

6.1 Baseline Selection

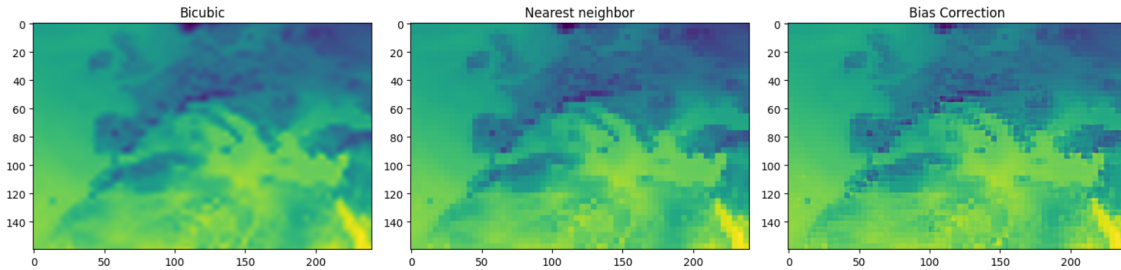
Table 6.1 presents the results of the comparison between the baseline methods described in Subchapter 5.3 of Methodology. The results indicate that bicubic interpolation outperforms all other methods across all evaluation metrics by a significant margin. This can be attributed to the fact that the original dataset was upsampled from $0.25^\circ \times 0.25^\circ$ to $0.5^\circ \times 0.5^\circ$ and finally to $1^\circ \times 1^\circ$ using bicubic interpolation, creating an inherent bias in favor of this method.

Furthermore, the bias correction applied to nearest neighbor interpolation does not appear to provide any meaningful improvement, as it simply adds a constant correction per cell. A more sophisticated approach—such as computing dynamic corrections based on seasonal or monthly variations—might yield better results.

	Bicubic	Nearest Neighbor	Bias Correction
MAE ↓	0.00607	0.00904	0.01382
MSE ↓	0.00016	0.00034	0.00092
PSNR ↑	37.8830	34.7709	30.3939
SSIM ↑	0.9493	0.9058	0.8257
Land MAE ↓	0.00944	0.01368	0.02150
Land MAE in °C ↓	0.60665	0.87871	1.38165

Table 6.1: Comparison between baseline methods.

In order to have a visual confirmation of the above results, in Figure 6.1 the downscaling results obtained using the three methods for a sample from the test dataset are presented. Notably, the bicubic interpolation appears to produce qualitatively better results than the other two methods, although the output is somewhat blurred.

**Figure 6.1:** Comparison between baseline methods.

Given these findings, bicubic interpolation will be used as the baseline method for comparison with the DL-based approaches throughout the analysis.

6.2 DL Model Architecture Selection

As described in Chapter 5 (Methodology), four architectures were compared to identify the most suitable model for this task: CAE, SRCNN, EDSR, and RCAN. All models were trained on the same dataset, with inputs at $1^\circ \times 1^\circ$ resolution and corresponding targets at $0.5^\circ \times 0.5^\circ$, for a maximum of 200 epochs. Early stopping

was enabled with a patience of 50 epochs, and the learning rate was scheduled to reduce after 40 epochs without improvement.

The CAE model reduced its learning rate at epoch 159 but continued training until the full 200 epochs without triggering early stopping. SRCNN experienced learning rate reductions at epochs 134 and 175 and eventually stopped training at epoch 185 due to stagnating performance. EDSR lowered its learning rate at epoch 169 and completed training at epoch 179. RCAN reduced its learning rate at epoch 172 and also trained until the maximum 200 epochs, without triggering early stopping. The full learning rate progression for all models throughout training is illustrated in Figure 6.2.

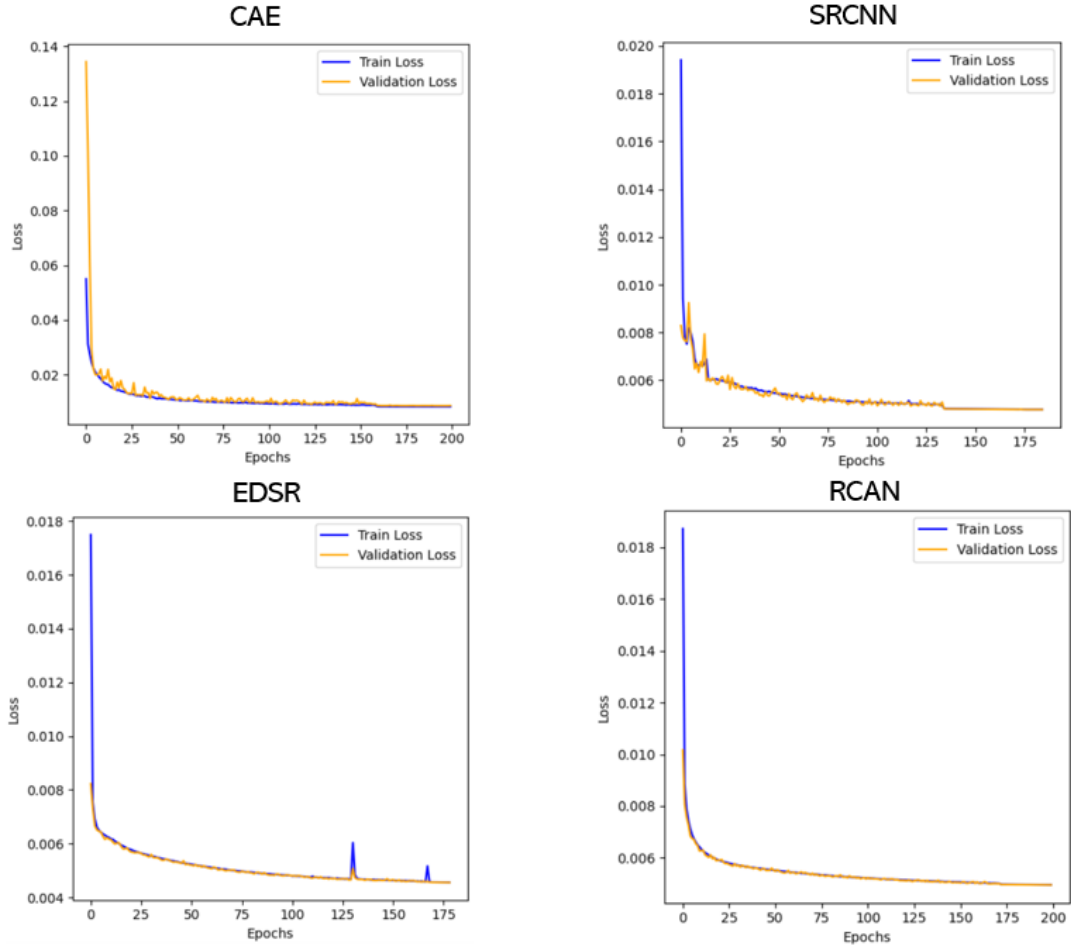


Figure 6.2: Comparison of training loss trajectories for the evaluated architectures.

All four models demonstrate successful convergence, with steadily decreasing training and validation losses. CAE and RCAN show smooth and stable learning

behavior throughout, while SRCNN and EDSR exhibit occasional fluctuations. Notably, EDSR achieves lower MAE values earlier in training compared to the others, indicating faster convergence to high-performance levels. While other models may require more data or longer training to reach similar accuracy, or potentially benefit from deeper or more complex architectures, EDSR strikes a strong balance between effectiveness and training efficiency, making it a practical and well-performing choice for this study.

The comparative results of the four models on the testing dataset are presented in the Table 6.2 and confirm that the architecture that achieves the best performance across evaluation metrics is EDSR, with SRCNN following closely behind. In terms of training speed, CAE is the fastest, while SRCNN has the fewest parameters, making it the most memory-efficient.

	CAE	SRCNN	EDSR	RCAN
MAE ↓	0.00865	0.00482	0.00466	0.00498
MSE ↓	0.00016	0.00008	0.00007	0.00009
PSNR ↑	38.0152	41.0553	41.4122	40.3840
SSIM ↑	0.9435	0.9733	0.9753	0.9726
Land MAE ↓	0.01134	0.00731	0.00703	0.00769
Land MAE in °C ↓	0.72874	0.46989	0.45191	0.49382
Training time per epoch ↓	6s	18s	13s	13s
Trainable params ↓	100,422,977	57,281	83,841	75,530
Non-trainable params ↓	1,344	0	0	0
Total params ↓	100,424,321	57,281	83,841	75,530

Table 6.2: Architecture comparison results.

Figure 6.3 presents the evaluation metrics and their progression throughout the training history of the EDSR model, which was identified as the best-performing architecture.

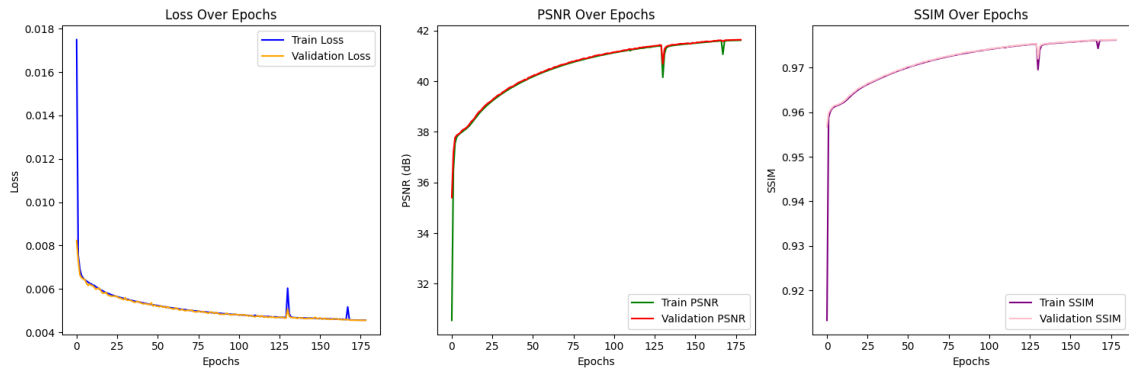


Figure 6.3: Training history of EDSR model.

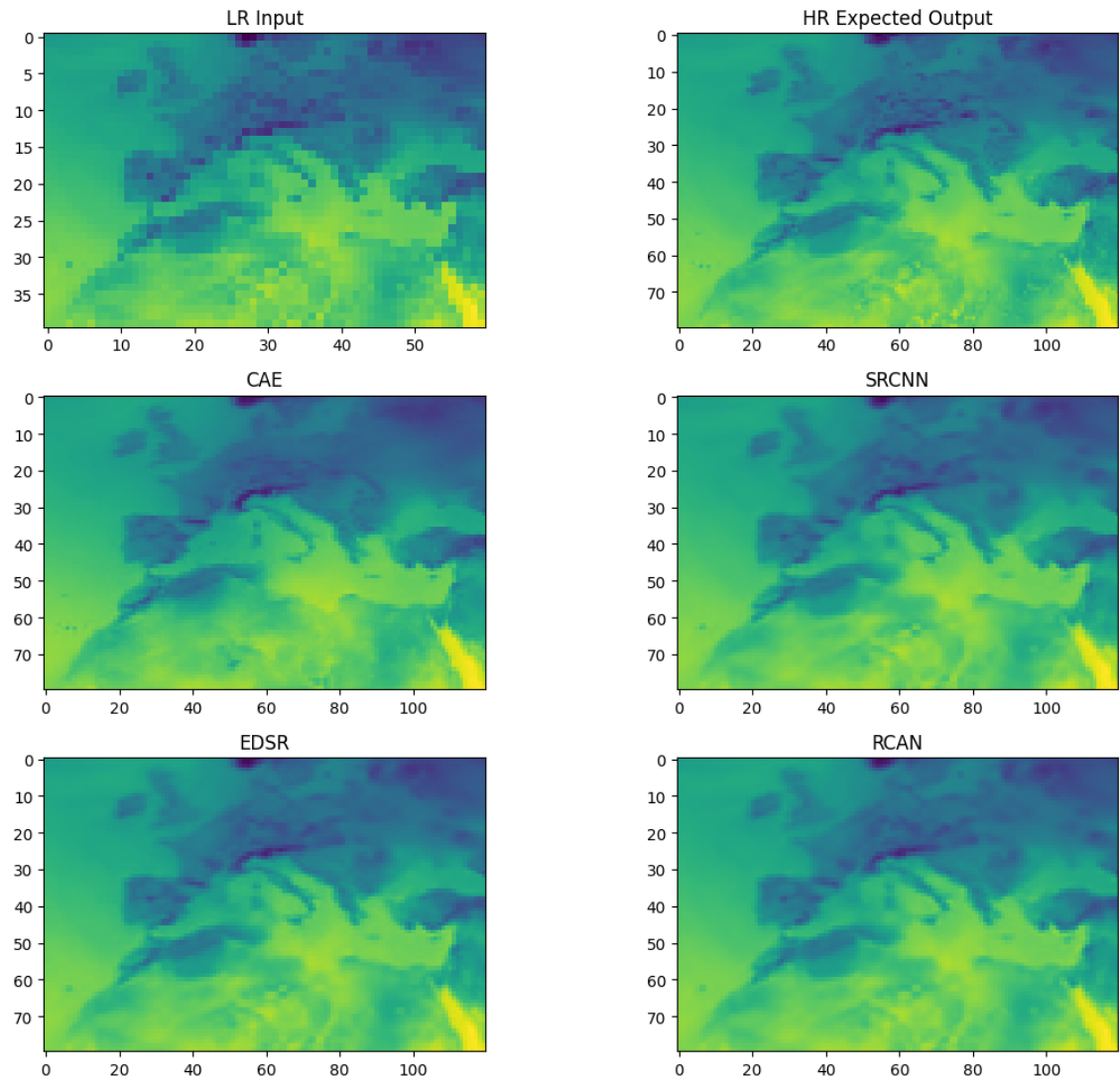


Figure 6.4: Comparison between the outputs of different models architecture zoomed in Europe.

Figure 6.4 illustrates an example output from each model, alongside the input data

and the expected high-resolution (HR) result for comparison. To highlight the differences more clearly, a zoomed-in view of the wider European region has been provided.

Another useful metric for comparing the models is the average MAE per pixel on the test dataset across the different architectures, as shown in Figure 6.5. In all cases, the error is more pronounced in the temperate zones and regions characterized by complex terrain morphology, where temperature variations are more significant. Notably, the CAE model exhibits higher MAE values across the landmass compared to the other models. Meanwhile, the RCAN model shows elevated error values in mountainous regions such as the Alps, Caucasus, Ethiopian Highlands, and the Himalayas, indicating challenges in accurately capturing temperature variations in these areas.

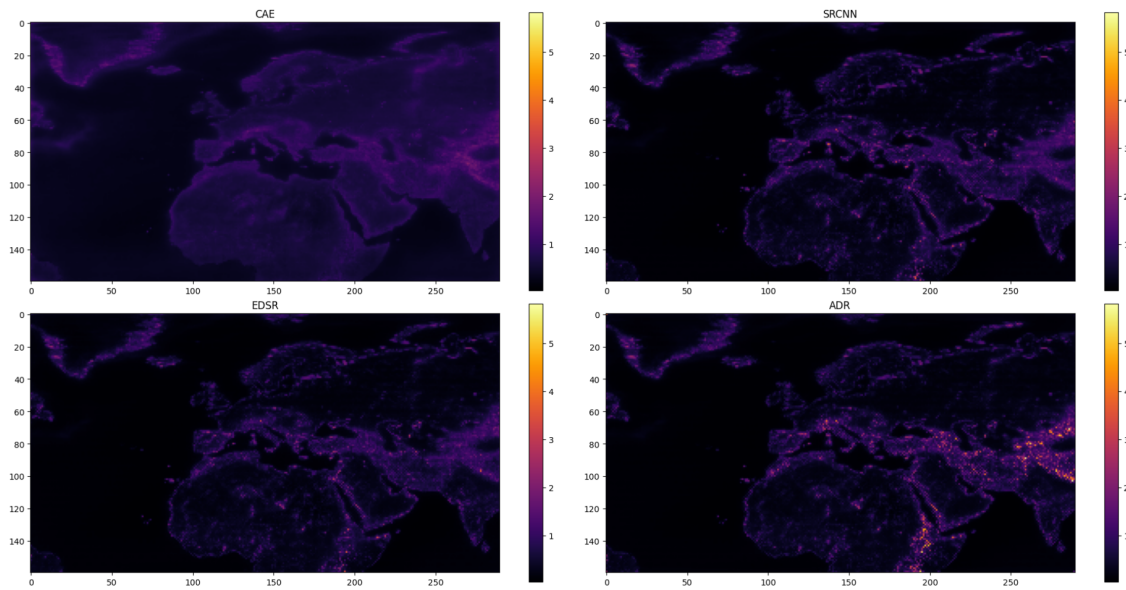


Figure 6.5: MAE of different models architecture.

Meanwhile, the average PSNR per pixel on the test dataset, shown in Figure 6.6, exhibits similar trends, further supporting the choice of EDSR as the most suitable architecture to proceed with.

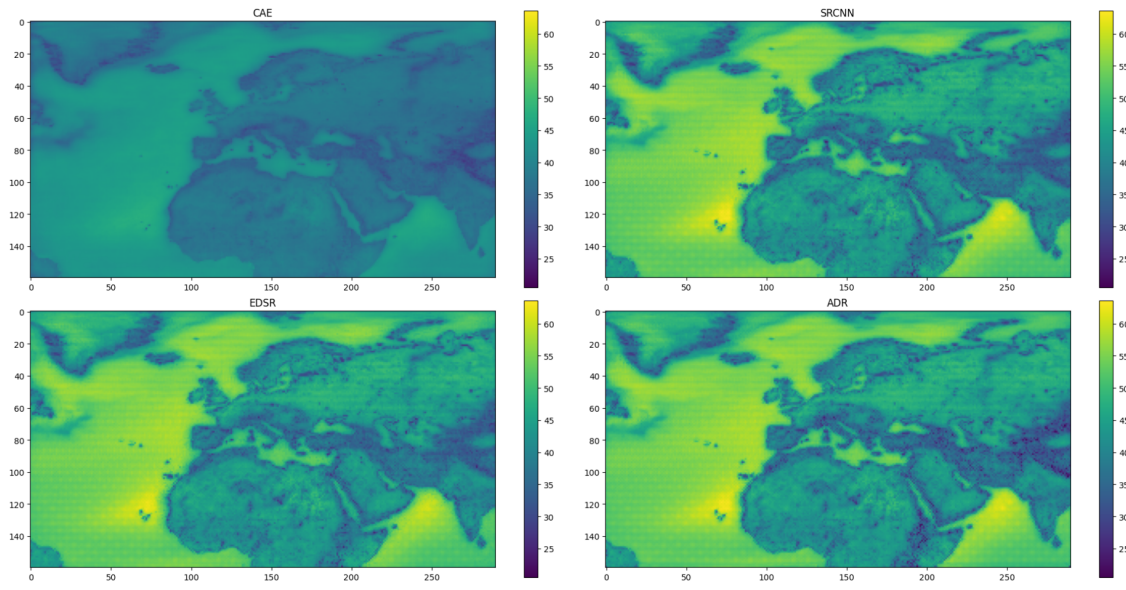


Figure 6.6: PSNR of different models architecture.

6.3 Comparison between seasonal and universal models

In this step, the goal was to investigate whether training a separate model for each season would lead to a significant improvement in downscaling performance. To this end, five EDSR models were trained to downscale from $1^\circ \times 1^\circ$ resolution to $0.5^\circ \times 0.5^\circ$: one universal model trained on the full dataset and four seasonal models, each trained exclusively on data from a single season. All models were trained for up to 1000 epochs with a batch size of 128, using early stopping with a patience of 150 epochs and learning rate reduction triggered after 100 epochs without improvement. The training history of the universal model is presented in Figure 6.7. In this case, the learning rate was reduced at epoch 351, and training concluded at epoch 401, suggesting that the default 1000-epoch limit may be excessive for datasets of this size.

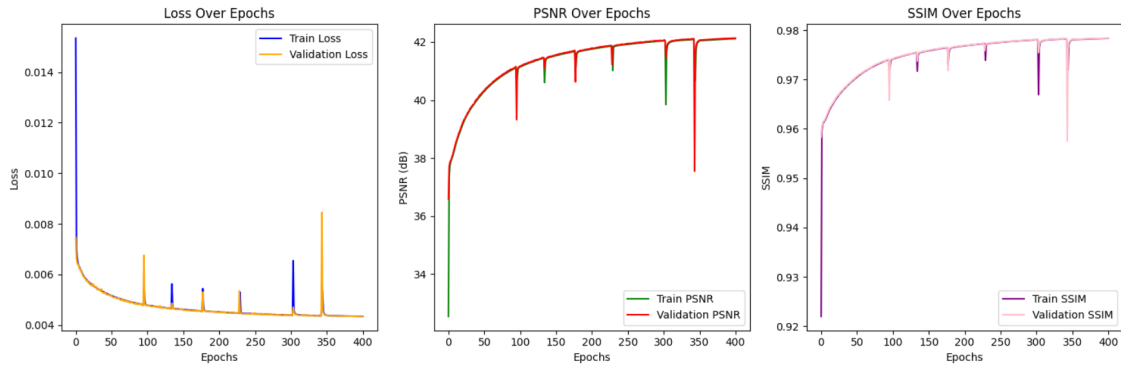


Figure 6.7: Training history of Universal EDSR model.

The training history of the universal EDSR model demonstrates a generally stable and effective learning process. The training and validation loss curves exhibit a consistent downward trend, indicating convergence. Occasional spikes in both curves are likely related to learning rate adjustments, but they do not appear to significantly impact the overall trend.

The seasonal models exhibit similarly good convergence behavior. Their training typically continues for approximately twice the number of epochs compared to the universal model. This is expected, as each seasonal model is trained on a subset of

the data—roughly one-quarter of the total training set—which requires more epochs to achieve comparable convergence. A comparison of their training loss curves is presented in Figure 6.8.

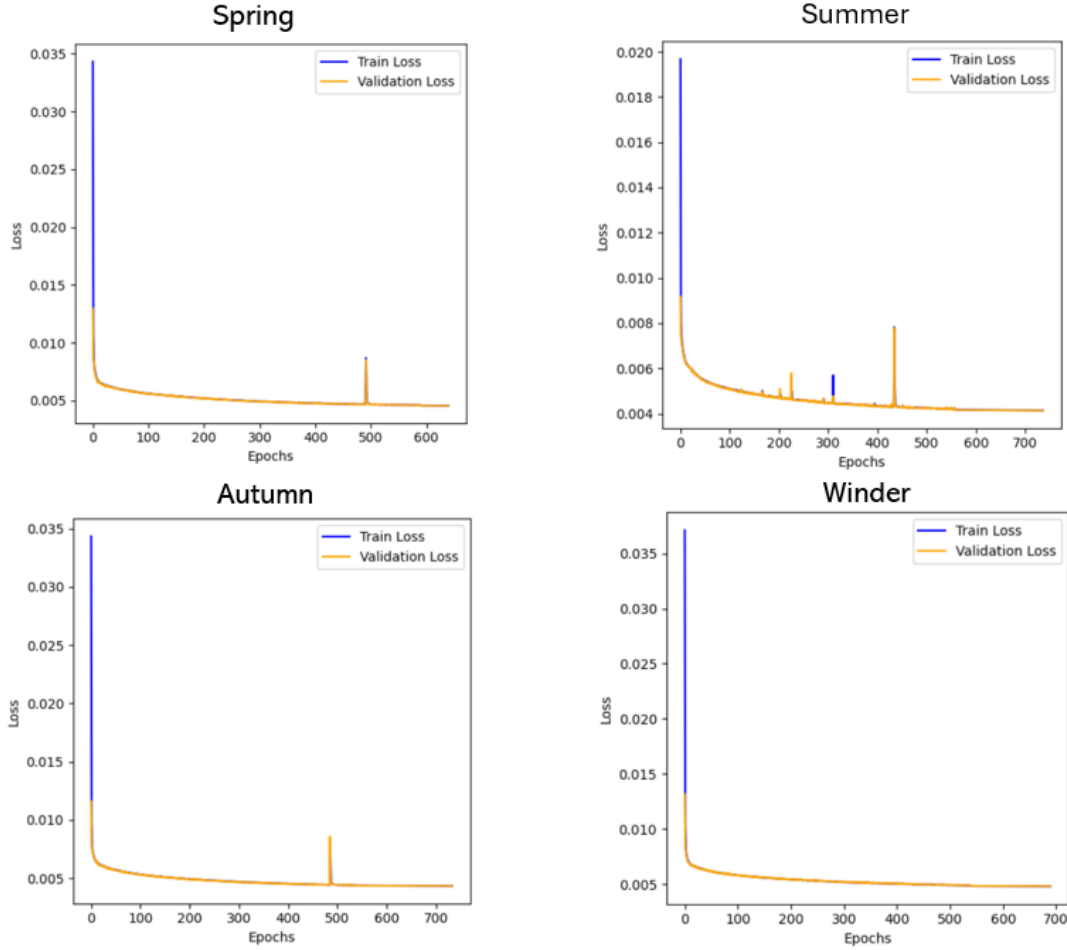


Figure 6.8: Comparison of training loss trajectories for each seasonal model.

Table 6.3 below presents a comparison of the performance metrics achieved by each method per season on the test dataset. Overall, the results show only minor differences between the seasonal and universal models. In fact, the universal model outperforms the seasonal ones in all seasons except summer. This can likely be attributed to two main factors:

1. The universal model is trained on a dataset that is four times larger than that of each seasonal model, providing a broader and more diverse learning base.
2. The study area spans a wide geographical range, encompassing both tropical and temperate climate zones. As a result, the conventional seasonal dis-

tinctions, based on temperate-zone definitions, may not meaningfully capture variations in tropical regions, which constitute a significant portion of the grid.

		Seasonal	Universal
Spring	MAE ↓	0.00464	0.00446
	MSE ↓	0.00007	0.00006
	PSNR ↑	41.5380	41.9422
	SSIM ↑	0.9747	0.9765
	Land MAE ↓	0.00706	0.00677
	Land MAE in °C ↓	0.45378	0.43484
Summer	MAE ↓	0.00417	0.00422
	MSE ↓	0.00005	0.00006
	PSNR ↑	42.6294	42.4040
	SSIM ↑	0.9784	0.9778
	Land MAE ↓	0.00620	0.00628
	Land MAE in °C ↓	0.39857	0.40343
Autumn	MAE ↓	0.00437	0.00427
	MSE ↓	0.00006	0.00006
	PSNR ↑	42.1571	42.3591
	SSIM ↑	0.9778	0.9788
	Land MAE ↓	0.00656	0.00640
	Land MAE in °C ↓	0.42152	0.41139
Winter	MAE ↓	0.00482	0.00478
	MSE ↓	0.00008	0.00008
	PSNR ↑	41.0601	41.1426
	SSIM ↑	0.9765	0.9770
	Land MAE ↓	0.00731	0.00722
	Land MAE in °C ↓	0.46974	0.46392

Table 6.3: Seasonal models vs Universal model results.

Based on the results of the above test, it was concluded that training separate models for each season adds unnecessary complexity to the methodology without offering a meaningful improvement in performance. This approach might be more

appropriate if the spatial domain were limited to a smaller, more climatically homogeneous area, where seasonal definitions are more relevant and consistent across the grid.

6.4 Comparison between 1-step and 2-steps downscaling

To evaluate whether downscaling yields better results when performed in one step or two, a comparison was conducted using the previously trained universal model (for $1^\circ \times 1^\circ$ to $0.5^\circ \times 0.5^\circ$ resolution), along with two additional models. The first of these new models was trained to further downscale from $0.5^\circ \times 0.5^\circ$ to $0.25^\circ \times 0.25^\circ$, effectively completing the second step in a two-stage approach. This model was trained for 1000 epochs with a batch size of 64, using early stopping with a patience of 150 epochs and ReduceLROnPlateau with a patience of 100.

The second model was trained to perform the full downscaling in a single step, from $1^\circ \times 1^\circ$ directly to $0.25^\circ \times 0.25^\circ$, using a batch size of 128 and the same early stopping and learning rate reduction settings.

As shown in Figure 6.9, this single-step model—despite the larger resolution jump—halted training at 411 epochs, nearly the same as the model trained from $1^\circ \times 1^\circ$ to $0.5^\circ \times 0.5^\circ$. This suggests that to effectively handle the increased complexity of a larger resolution jump, the model may need to be deeper and more expressive in order to reach a lower loss.

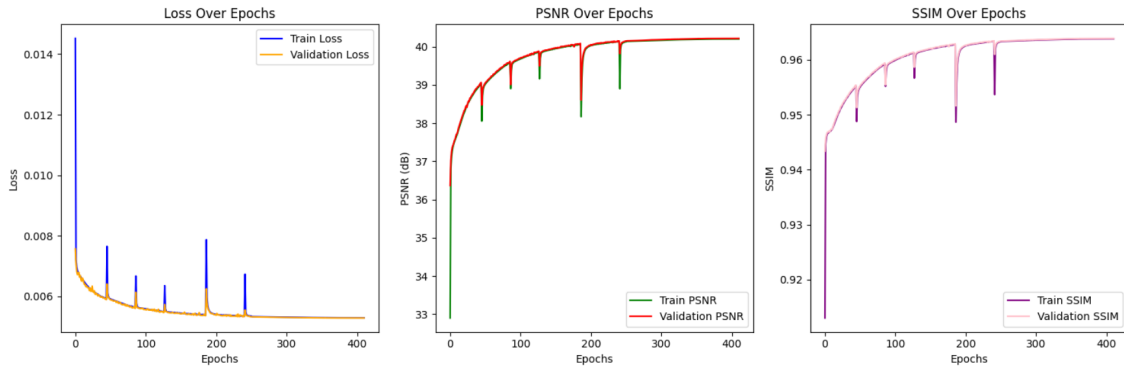


Figure 6.9: Training history of 1° to 0.25° EDSR model.

Figure 6.10 below presents the visual results of downscaling using the two ap-

proaches for a sample from the test dataset. The displayed region is a zoomed-in view over Greece, chosen to make differences in spatial detail more apparent.

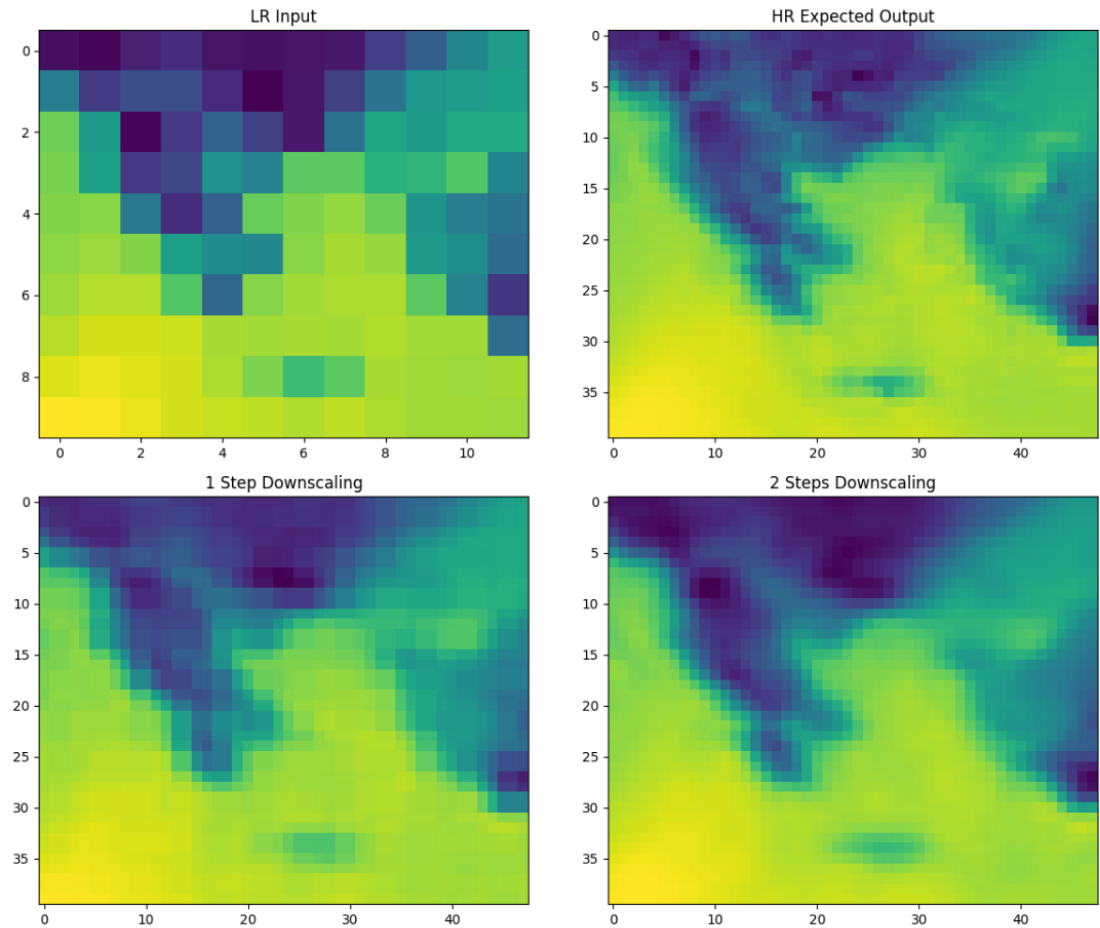


Figure 6.10: Comparison between the outputs of 1 and 2 steps downscaling approach.

Following the visual comparison, Table 6.4 below summarizes the performance metrics for both downscaling approaches, along with the baseline, to highlight the improvements achieved. It is evident that the 2-steps approach consistently outperforms the 1-step method across all metrics. Notably, for the land-area MAE in °C, which is the primary focus of this study, the 2-steps method shows an average improvement of approximately 0.1°C.

	Baseline	1 Step	2 Steps
MAE ↓	0.00607	0.00533	0.00431
MSE ↓	0.00016	0.00010	0.00007
PSNR ↑	37.8830	40.1433	41.8425
SSIM ↑	0.9493	0.9633	0.9721
Land MAE ↓	0.00944	0.00800	0.00652
Land MAE in °C ↓	0.60665	0.51400	0.41916
Training time per epoch ↓	-	23s	13s + 48s
Trainable params ↓	-	83,841	83,841 + 83,841
Non-trainable params ↓	-	0	0
Total params ↓	-	83,841	83,841 + 83,841

Table 6.4: Comparison of downscaling methods.

Finally, the MAE (in °C) across the entire study area (Figure 6.11) further underscores the superiority of the two-step downscaling approach. Specifically, the 2-steps method achieved a mean per-pixel MAE of 0.2768°C and a maximum MAE of 4.0443°C, whereas the 1-step method recorded a higher mean of 0.3425°C and a significantly larger maximum error of 9.4054°C. This key comparison demonstrates not only the overall improved accuracy of the 2-steps approach but also its enhanced reliability in handling extreme cases, where even the largest errors remain closer to the true values.

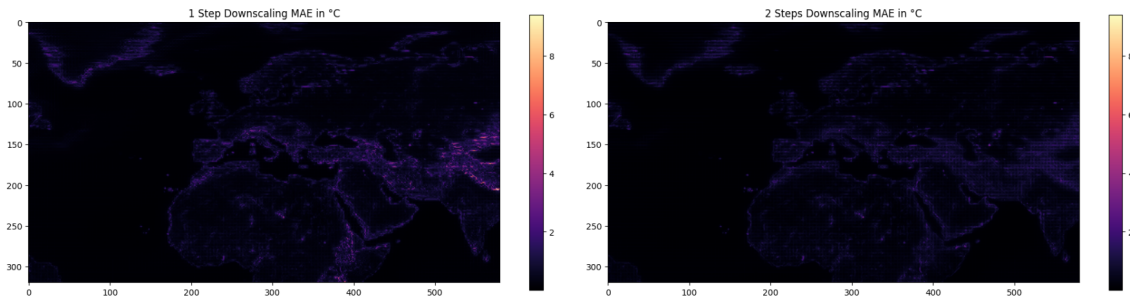


Figure 6.11: Per-pixel MAE in °C for one and two steps downscaling method.

Therefore, based on the results above, it is confirmed that a gradual, multi-step

downscaling approach can outperform direct downscaling. This is likely because it allows the model to better capture and adapt to the different physical processes and influencing factors present at each intermediate spatial scale.

6.5 Comparison between methods to incorporate DEM data

As outlined in Subchapter 5.9 of the Methodology, two strategies were tested for integrating DEM data into the model training process:

1. Incorporating elevation data as a second input channel alongside the primary input.
2. Injecting the elevation data after the upsampling stage, as a second channel to the intermediate output.

These two approaches were initially tested separately using the $1^\circ \times 1^\circ$ to $0.5^\circ \times 0.5^\circ$ resolution step model. Training was conducted for 1000 epochs with a batch size of 128, employing early stopping with a patience of 150 epochs and learning rate reduction after 100 epochs. The training history of the 2nd input channel method is shown in Figure 6.12, demonstrating smooth and stable convergence. This method also achieved a minor improvement in MAE on the test dataset, an decrease of 0.0004 compared to the model without DEM integration.

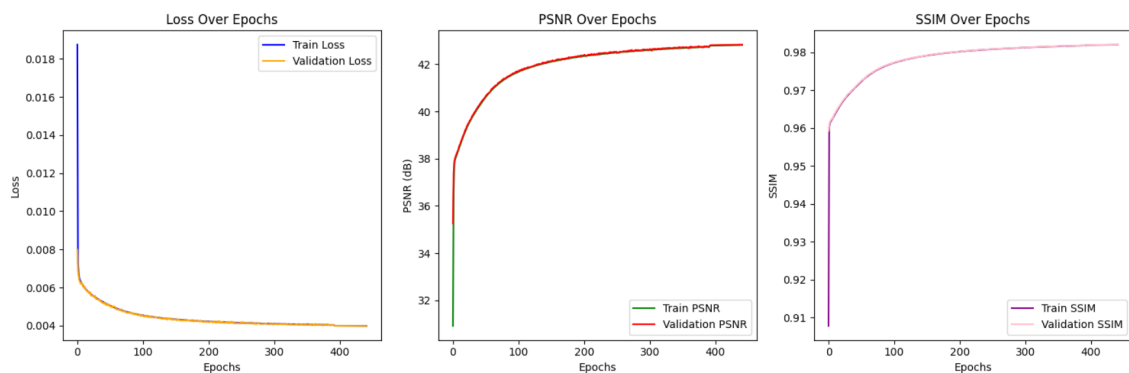


Figure 6.12: Training history of 2nd Input Channel DEM data EDSR model.

The second approach (Post-upsampling fusion method) was also evaluated using the $1^\circ \times 1^\circ$ to $0.5^\circ \times 0.5^\circ$ downscaling model, under the same training conditions. The

training history for this method is illustrated in Figure 6.13. Training and validation losses drop smoothly overall, but sudden spikes occur around epochs 80 and 260. These do not correspond to the learning rate reduction (which happened at epoch 271), so other likely causes include checkpoint reloads or instability during batch normalization updates. Despite these artifacts, the model quickly corrected and, in the end, achieved an almost equal improvement of MAE with the 1st method.

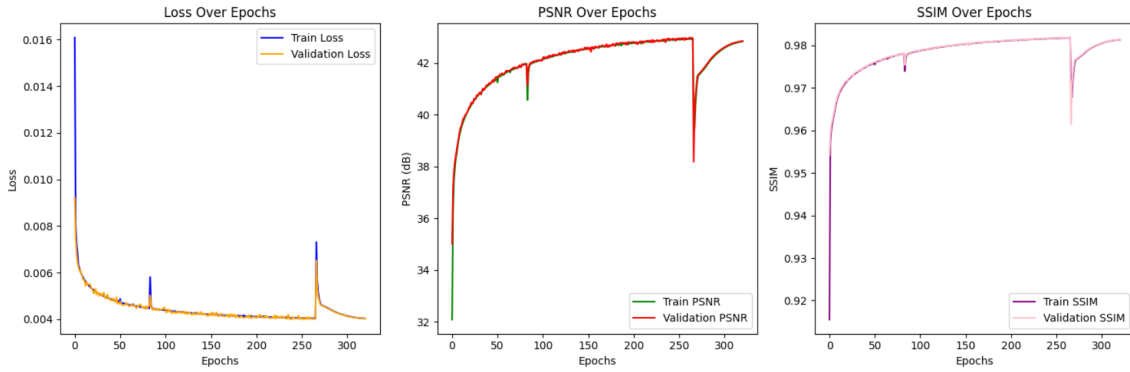


Figure 6.13: Training history of Last Step Fusion DEM data EDSR model.

To take advantage of the strengths of both previous strategies, a third model was developed that integrates both DEM augmentation approaches within a unified architecture. This model was trained under identical conditions to ensure a fair comparison. As shown in the training history (Figure 6.14), the spikes observed in the post-upsampling method were significantly mitigated, indicating a more stable and consistent convergence process. Notably, it achieved double the improvement in MAE compared to either of the individual methods, confirming the complementary benefits of combining DEM data both pre- and post-upsampling.

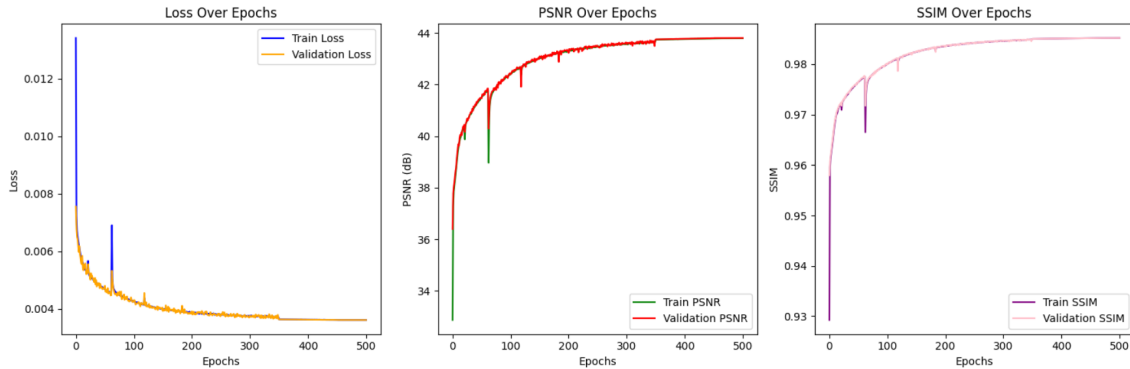


Figure 6.14: Training history of both methods DEM data EDSR model.

6.5 : Comparison between methods to incorporate DEM data

Table 6.5 below presents the detailed results of the previously described experiments, allowing for direct comparison with the baseline bicubic interpolation method. Notably, the combined DEM augmentation approach leads to a substantial improvement, achieving an average reduction of 0.07°C in Land MAE, which is particularly meaningful in climate data analysis. Additionally, the PSNR increases by approximately 2dB, indicating a marked enhancement in the reconstruction quality and fidelity of the downscaled outputs. These results highlight the effectiveness of incorporating elevation data both before and after upsampling, as it enables the model to better capture spatial variability and terrain-influenced climate patterns.

	No DEM	2nd Channel	Fusion	Combination
MAE ↓	0.00443	0.00409	0.00408	0.00364
MSE ↓	0.00006	0.00005	0.00005	0.00004
PSNR ↑	41.9274	42.6073	42.6945	43.7255
SSIM ↑	0.9775	0.9812	0.9807	0.9850
Land MAE ↓	0.00667	0.00607	0.00615	0.00546
Land MAE in $^{\circ}\text{C}$ ↓	0.42867	0.38988	0.39514	0.35070
Training time per epoch ↓	13s	13s	18s	19s
Trainable params ↓	83,841	84,129	97,857	98,145
Non-trainable params ↓	0	0	0	0
Total params ↓	83,841	84,129	97,857	98,145

Table 6.5: Comparison of different methods to incorporate DEM data.

After demonstrating the effectiveness of incorporating DEM data and selecting the optimal integration approach, the $0.5^{\circ}\times 0.5^{\circ}$ to $0.25^{\circ}\times 0.25^{\circ}$ resolution step model was also trained with the same architecture. The same training conditions were applied, with the exception of a reduced batch size of 16 due to limited computational

resources. The results of this final methodology are presented in the following subsection.

6.6 Impact of DEM Data on Performance

The results of the final proposed methodology of this study, consisting of two sequential models (one for the $1^\circ \times 1^\circ$ to $0.5^\circ \times 0.5^\circ$ step and another for the $0.5^\circ \times 0.5^\circ$ to $0.25^\circ \times 0.25^\circ$ step) both incorporating DEM data augmentation, are summarized in Table 6.6 below. This approach is compared against the corresponding models without DEM augmentation, as well as against the baseline bicubic interpolation method, to support the final conclusions of the study.

	Baseline	No DEM	DEM-Augmented
MAE ↓	0.00607	0.00431	0.00354
MSE ↓	0.00016	0.00007	0.00004
PSNR ↑	37.8830	41.8425	43.6232
SSIM ↑	0.9493	0.9721	0.9810
Land MAE ↓	0.00944	0.00652	0.00534
Land MAE in °C ↓	0.60665	0.41916	0.34326
Training time per epoch ↓	-	13s + 48s	18s + 79s
Trainable params ↓	-	83,841 + 83,841	98,145 + 98,145
Non-trainable params ↓	-	0	0
Total params ↓	-	83,841 + 83,841	98,145 + 98,145

Table 6.6: Comparison of different methods to incorporate DEM data.

It is important to highlight that the final proposed method reduces the baseline MAE by approximately 50%, while also improving the PSNR by around 6dB. Furthermore, regarding the land MAE in degrees Celsius, which is the most critical metric in this context—the final model achieves an error of 0.34°C . This represents

nearly a 50% reduction compared to the baseline methods and an improvement of about 0.1°C over the version without DEM augmentation.

The violin plots below (Figure 6.15) illustrate the distribution of per-pixel evaluation metrics on the test dataset for the three compared approaches: baseline bicubic interpolation, the model without DEM integration, and the final DEM-augmented model.

A key observation is that the DEM-augmented method not only achieves the best central performance in all metrics, but also exhibits a much tighter distribution, indicating improved consistency across the entire test set. Most notably:

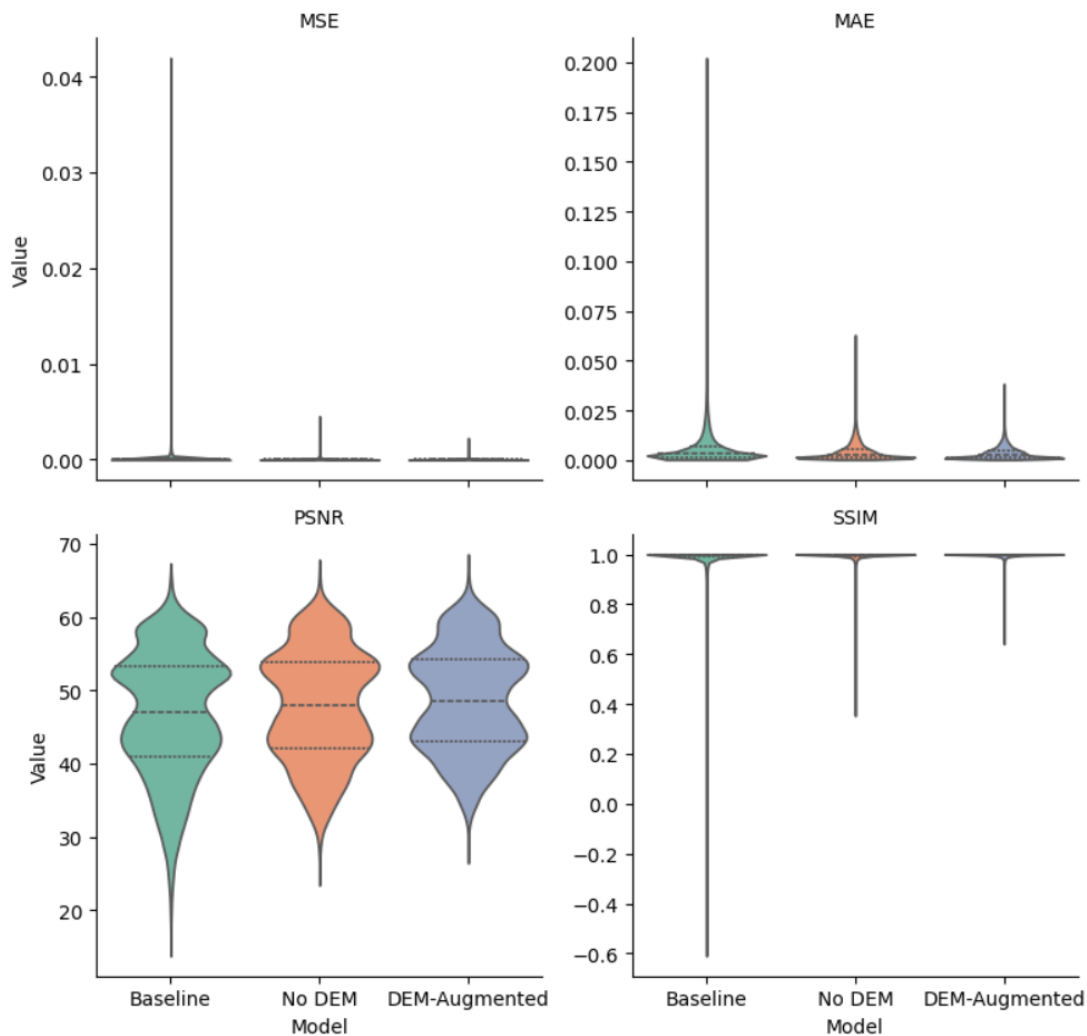


Figure 6.15: Per-Pixel metrics comparison between Baseline, No DEM and DEM-Augmented models.

- **MSE and MAE:** The DEM-augmented model shows significantly reduced

spread and outliers compared to the other two methods, highlighting fewer extreme errors.

- **PSNR:** The PSNR values are consistently higher in the DEM-augmented case, with a noticeably narrower distribution and a higher median, suggesting more reliable reconstruction quality.
- **SSIM:** While all models exhibit high SSIM scores, the DEM-augmented model achieves nearly perfect structural similarity across the board, with minimal variance.

The difference in per-pixel MAE in $^{\circ}\text{C}$ can also be visually assessed from Figure 6.15. In the first image, corresponding to the baseline method, large errors are clearly evident. These errors are noticeably reduced in the second image, representing the model without DEM data. In the third image, generated by the DEM-augmented approach, the errors are almost entirely eliminated, indicating a substantial improvement in prediction accuracy.

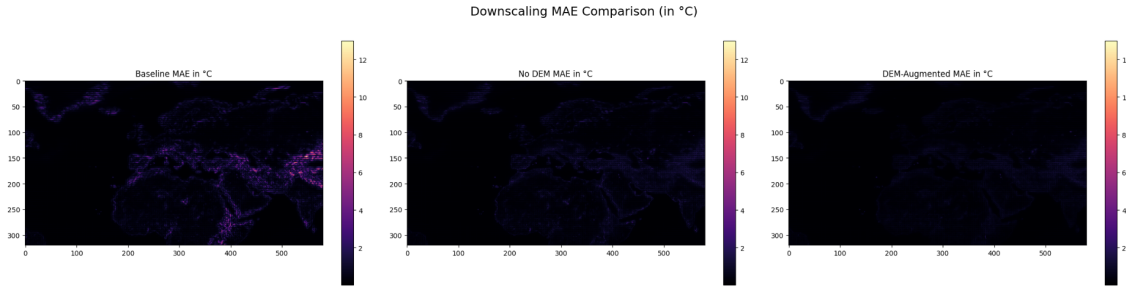


Figure 6.16: Per-Pixel MAE in $^{\circ}\text{C}$ of Baseline, No DEM and DEM-Augmented models.

Specifically, the per-pixel MAE plots in $^{\circ}\text{C}$ (Figure 6.17) reveal a substantial improvement with the DEM-augmented approach. The baseline method exhibits a mean error of 0.3902°C and a maximum error of 12.9912°C . The No DEM model reduces these values to 0.2768°C mean and 4.0443°C max. The DEM-augmented model further improves the results, achieving a mean error of just 0.2273°C and a maximum of 2.4681°C . Qualitatively, this indicates that even in the worst-case scenarios, the DEM-augmented model deviates from the true value by only about 2.5°C on average—significantly lower than the roughly 4°C deviation seen in the No DEM variant.

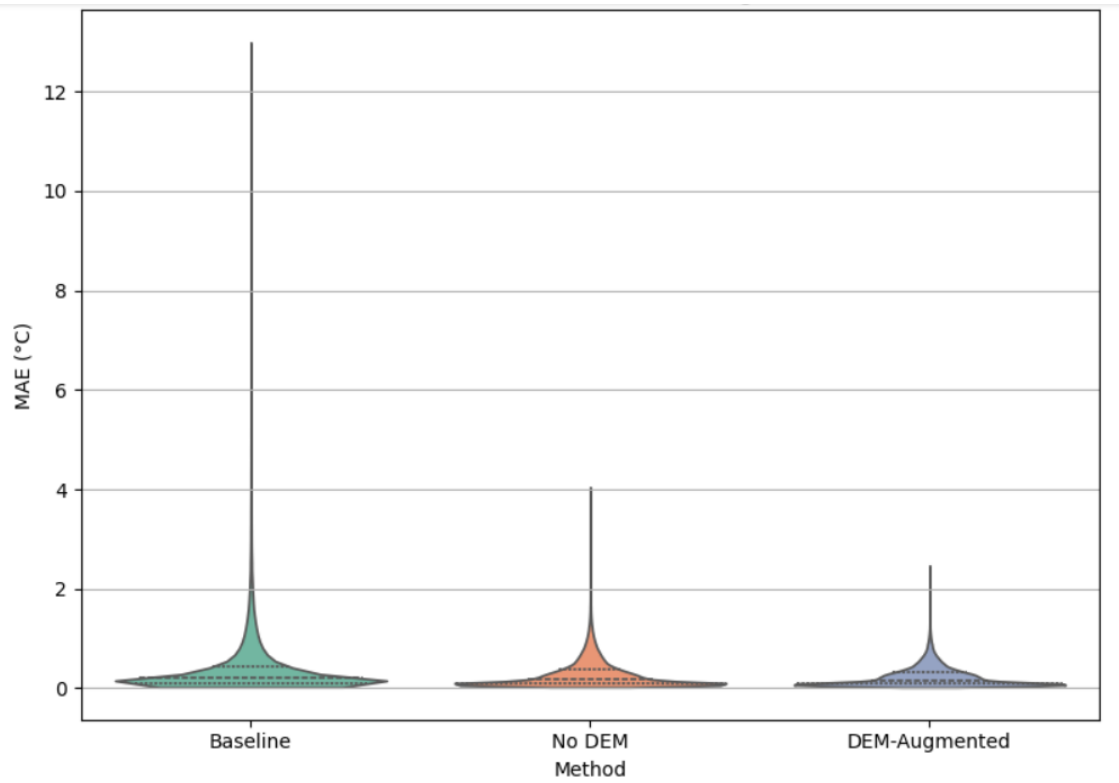


Figure 6.17: Per-Pixel MAE in °C in violin plots.

To provide a clearer view of the MAE in °C, Figure 6.18 presents the same data as Figure 6.16 but excludes the baseline, which contained large outliers. This allows for a better comparison between the No DEM and DEM methods. As shown in the left image (No DEM model), the more intense colors indicate larger errors compared to the DEM-Augmented model. In fact, the No DEM model contains several pixels with significantly large error outliers, whereas the final method has far fewer such pixels and with smaller error values.

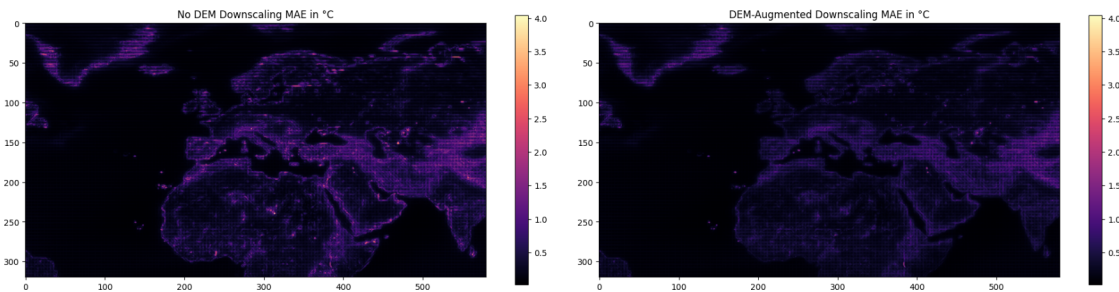


Figure 6.18: Per-Pixel MAE in °C of No DEM and DEM-Augmented models.

Finally, Figures 6.19 and 6.20 below present results from a test dataset instance

for the No DEM and DEM-Augmented models, alongside the LR input and the expected HR output for comparison. In Figure 6.19, the output is zoomed to the broader European region to highlight differences more clearly, while in Figure 6.20, the output is focused on the Greek region.

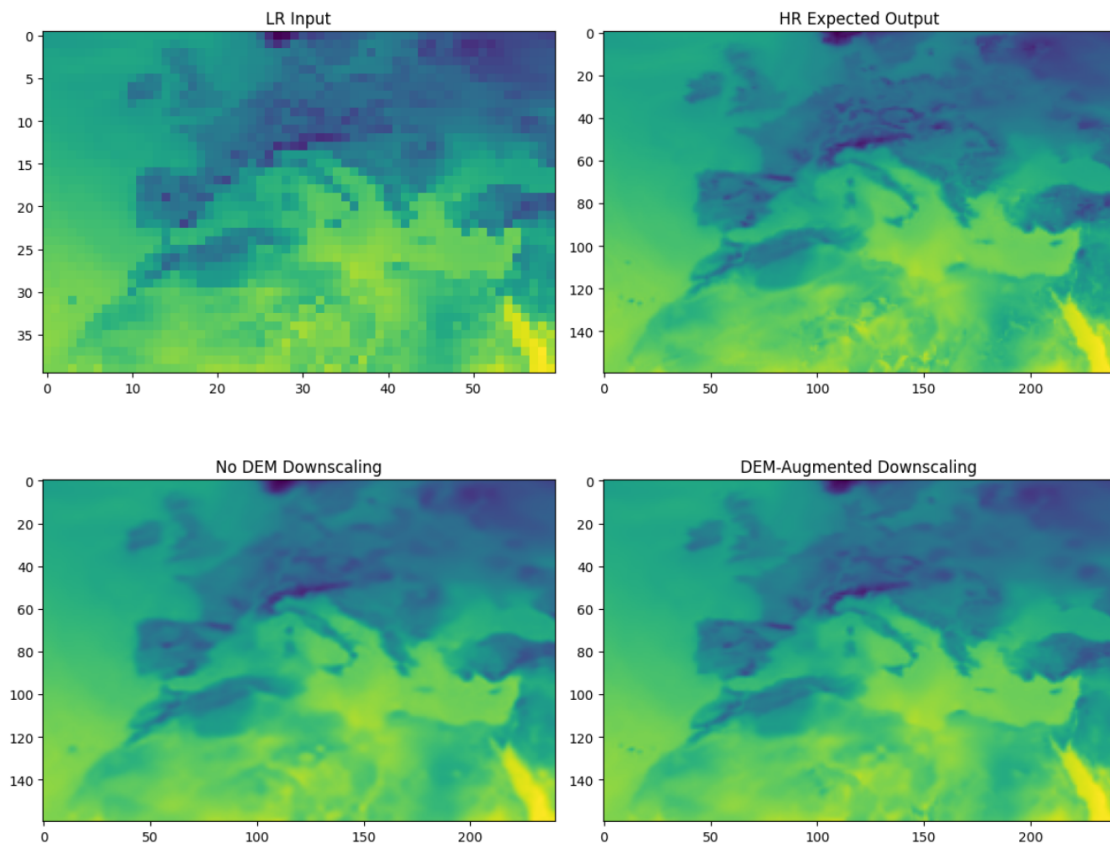


Figure 6.19: Comparison between the outputs of No DEM and DEM-Augmented approaches zoomed in Europe.

In Europe, elevation data have helped to more clearly visualize temperature differences in the Carpathians, Pyrenees, and the Giant Mountains (Karkonosze) in Poland. Similarly, in the Greek region, the island of Crete is more distinctly defined, while temperature variations are more noticeable in the Pindus and Olympus mountain ranges.

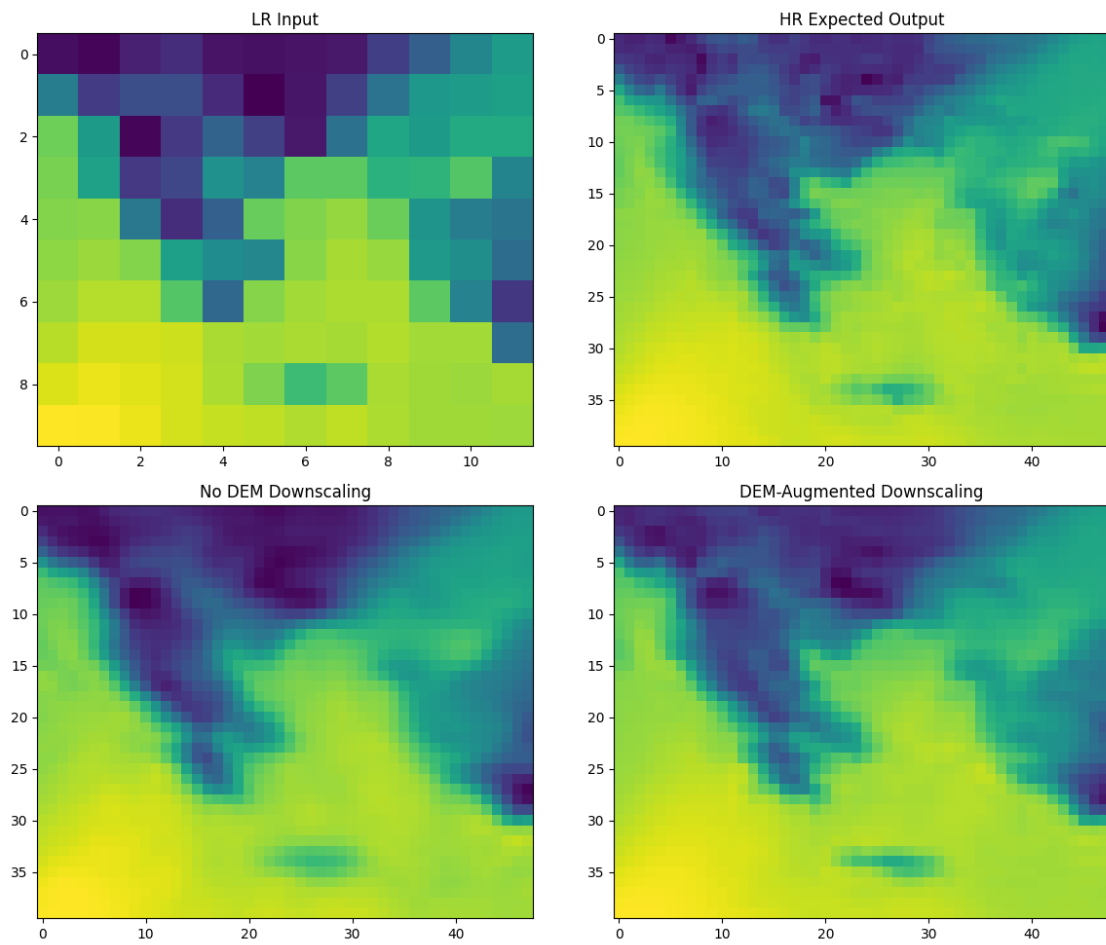


Figure 6.20: Comparison between the outputs of No DEM and DEM-Augmented approached zoomed in Greece.

Chapter 7

Conclusions and Future Work

This study presented an effective and lightweight method for downscaling climate data, capable of doubling spatial resolution while maintaining a low average error of approximately 0.2°C per cell. The core innovation lies in two complementary enhancements: a gradual two-step downscaling process, and the integration of geospatial information, particularly elevation, into the DL models. These improvements significantly enhance the accuracy of the downscaled outputs, particularly in complex terrains such as mountainous regions, where traditional methods tend to perform poorly.

The study shows that these relatively simple yet targeted interventions shift the problem from a generic SISR task toward a domain-informed climate application. By doing so, the models are better able to capture spatial variability in temperature that is influenced by underlying geographic and topographic features. This not only improves performance but also increases the physical interpretability of the results, an important consideration in climate modeling.

Additionally, the approach is designed to be model-agnostic. It relies solely on ERA5 reanalysis data, which are widely available and standardized. This ensures that the proposed framework can be applied to a variety of climate or meteorological models, provided that the input data are reprojected to match ERA5's spatial and coordinate system specifications. As such, the methodology has broad applicability and can serve as a flexible tool in different regional or global downscaling tasks.

While the proposed method has demonstrated promising results in improving the spatial resolution of temperature data using DL and geospatial augmentation, several directions remain open for further exploration. The following key points are suggested for further investigation:

1. **Extension to other climate variables:** Temperature is only one of the key parameters shaping the climate. Future work could explore the incorporation of additional variables such as precipitation and wind, which are critical to understanding and modeling climate systems more comprehensively.
2. **Multi-variable modeling:** Investigating whether modeling these variables simultaneously, rather than independently, can improve performance is another promising direction. Since there are physical and statistical dependencies between variables (e.g., temperature and wind affecting evaporation or precipitation), a multi-target learning approach may lead to more accurate and physically consistent results.
3. **Finer-scale downscaling using higher-resolution reanalysis datasets:** The current method could be further enhanced by using reanalysis datasets with higher spatial resolution, such as ERA5-Land (with a 9km horizontal resolution) or CERRA-Land (with 5.5km resolution). These datasets could enable an additional level of downscaling, capturing finer local variability, especially in regions with complex topography. At this scale, land use and land cover become increasingly influential factors affecting surface temperature. Therefore, incorporating land use data as an additional input feature could further improve the model’s accuracy by accounting for urban heat islands, vegetation effects, and surface heterogeneity.
4. **Model architecture and training optimizations:** Additional performance gains may be achievable through further optimization of the model architecture and the size and diversity of the training dataset. With adequate computational resources, the best-performing EDSR model could be deepened to increase its capacity to learn from complex patterns. Similarly, the RCAN architecture, which already showed promising results, could be revisited and

potentially scaled up. As an improved successor to EDSR, a deeper RCAN may even outperform the current best model when adequately trained on large-scale datasets.

These directions highlight the potential for both methodological and practical improvements in climate data downscaling. By expanding the range of input variables, leveraging higher-resolution datasets, and refining model architectures, future research can build upon the current framework to produce even more accurate, physically meaningful, and versatile climate projections.

References

- [1] Liuyi Chen, Bocheng Han, Xuesong Wang, Jiazhen Zhao, Wenke Yang, and Zhengyi Yang. Machine learning methods in weather and climate applications: A survey. *Applied Sciences*, 13(21):12019, November 2023.
- [2] Yi Sun. Exploring deep learning strategies and prospective developments. *Transactions on Computer Science and Intelligent Systems Research*, 5:470–475, August 2024.
- [3] Copernicus Climate Change Service. Era5 hourly data on pressure levels from 1940 to present, 2018.
- [4] Muhammad Waqas, Usa Wannasingha Humphries, Bunthid Chueasa, and Angkool Wangwongchai. Artificial intelligence and numerical weather prediction models: A technical survey. *Natural Hazards Research*, 2024.
- [5] O.S. Ci. Image super resolution: A comparison between interpolation & deep learning-based techniques. <https://shorturl.at/ucUMU>, 2023. Accessed: 2025-06-05.
- [6] Dor Bank, Noam Koenigstein, and Raja Giryes. *Autoencoders*, pages 353–374. Springer International Publishing, Cham, 2023.
- [7] Chao Dong, Chen Change Loy, and Xiaoou Tang. Accelerating the super-resolution convolutional neural network, 2016.
- [8] Bee Lim, Sanghyun Son, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee. Enhanced deep residual networks for single image super-resolution, July 2017.

- [9] Yulun Zhang, Kunpeng Li, Kai Li, Lichen Wang, Bineng Zhong, and Yun Fu. Image super-resolution using very deep residual channel attention networks. 2018.
- [10] Mous Chahine Steve Graham, Claire Parkinson. Weather forecasting through the ages, 2002.
- [11] Stelios Karozis, Iraklis A. Klampanos, Athanasios Sfetsos, and Diamando Vlachogiannis. A deep learning approach for spatial error correction of numerical seasonal weather prediction simulation data. *Big Earth Data*, 7(2):231–250, February 2023.
- [12] Yoo-Geun Ham, Jeong-Hwan Kim, and Jing-Jia Luo. Deep learning for multi-year enso forecasts. *Nature*, 573(7775):568–572, September 2019.
- [13] Jerald A Brotzge, Don Berchoff, DaNa L Carlis, Frederick H Carr, Rachel Hogan Carr, Jordan J Gerth, Brian D Gross, Thomas M Hamill, Sue Ellen Haupt, Neil Jacobs, et al. Challenges and opportunities in numerical weather prediction. *Bulletin of the American Meteorological Society*, 104(3):E698–E705, 2023.
- [14] Frauke Feser, Burkhardt Rockel, Hans von Storch, Jörg Winterfeldt, and Matthias Zahn. Regional climate models add value to global model data: a review and selected examples. *Bulletin of the American Meteorological Society*, 92(9):1181–1192, 2011.
- [15] Alex Chang, Hugo Lee, Rong Fu, and Qi Tang. A seamless approach for evaluating climate models across spatial scales. *Frontiers in Earth Science*, 11:1245815, 2023.
- [16] Markku Rummukainen. Added value in regional climate modeling. *WIREs Climate Change*, 7(1):145–159, Nov 2015.
- [17] William J Gutowski Jr, Paul Aaron Ullrich, Alex Hall, L Ruby Leung, Travis Allen O’Brien, Christina M Patricola, Raymond W Arritt, Melissa S Bukovsky, Katherine V Calvin, Zhe Feng, et al. The ongoing need for high-resolution regional climate models: Process understanding and stakeholder in-

- formation. *Bulletin of the American Meteorological Society*, 101(5):E664–E683, 2020.
- [18] Marijn Van Der Meer, Sophie de Roda Husman, and Stef Lhermitte. Deep learning regional climate model emulators: A comparison of two downscaling training frameworks. *Journal of Advances in Modeling Earth Systems*, 15(6):e2022MS003593, 2023.
- [19] Jose González-Abad and José Manuel Gutiérrez. Are deep learning methods suitable for downscaling global climate projections? review and intercomparison of existing models. *arXiv preprint arXiv:2411.05850*, 2024.
- [20] Ritesh Kolte Abhishek Arora. Single image super-resolution. 2011.
- [21] Shalini Dubey, Prof. Pankaj Sahu, and Prof. Surya Bazal. Literature review on single image super resolution. *International Journal of Trend in Scientific Research and Development*, Volume-2(Issue-5):2485–2490, August 2018.
- [22] Qizhou Chen and Qing Shao. Single image super-resolution based on trainable feature matching attention network. *Pattern Recognition*, 149:110289, May 2024.
- [23] Karansingh Chauhan, Shail Nimish Patel, Malaram Kumhar, Jitendra Bhatia, Sudeep Tanwar, Innocent Ewean Davidson, Thokozile F. Mazibuko, and Ravi Sharma. Deep learning-based single-image super-resolution: A comprehensive review. *IEEE Access*, 11:21811–21830, 2023.
- [24] Kevin E. Trenberth, Toshio Koike, and Kazutoshi Onogi. Progress and prospects for reanalysis for weather and climate. *Eos, Transactions American Geophysical Union*, 89(26):234–235, June 2008.
- [25] Copernicus Climate Change Service. Reanalysis q&as, 2023. Accessed: 2025-06-03.
- [26] Copernicus Climate Change Service. Era5-land hourly data from 1950 to present, 2019.
- [27] Copernicus Climate Change Service. Cerra sub-daily regional reanalysis data for europe on single levels from 1984 to present, 2021.

- [28] Rafael C. Gonzalez and Richard E. Woods. *Digital Image Processing*. Pearson, 4th edition, 2018.
- [29] Mukhriddin Arabboev, Shohruh Begmatov, Mokhirjon Rikhsivoev, Khabibullo Nosirov, and Saidakmal Saydiakbarov. A comprehensive review of image super-resolution metrics: classical and ai-based approaches. *Acta IMEKO*, 13(1):1–8, March 2024.
- [30] Alexander M. Mood, Franklin A. Graybill, and Duane C. Boes. *Introduction to the Theory of Statistics*. McGraw-Hill, New York, 3rd edition, 1974.
- [31] Patrick Schneider and Fatos Xhafa. Anomaly detection, 2022.
- [32] Rafael C. Gonzalez and Richard E. Woods. *Digital Image Processing*. Prentice Hall, Upper Saddle River, NJ, 3rd edition, 2008.
- [33] Zhou Wang, A.C. Bovik, H.R. Sheikh, and E.P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.
- [34] Shutong Ye, Shengyu Zhao, Yaocong Hu, and Chao Xie. Single-image super-resolution challenges: A brief review. *Electronics*, 12(13):2975, July 2023.
- [35] Claude E. Duchon. Lanczos filtering in one and two dimensions. *Journal of Applied Meteorology and Climatology*, 18(8):1016 – 1022, 1979.
- [36] Antonio Marquina and Stanley J. Osher. Image super-resolution by tv-regularization and bregman iteration. *Journal of Scientific Computing*, 37(3):367–382, July 2008.
- [37] Jian Sun, Zongben Xu, and Heung-Yeung Shum. Image super-resolution using gradient profile prior. In *2008 IEEE Conference on Computer Vision and Pattern Recognition*, pages 1–8, 2008.
- [38] W.T. Freeman, T.R. Jones, and E.C. Pasztor. Example-based super-resolution. *IEEE Computer Graphics and Applications*, 22(2):56–65, 2002.
- [39] Hong Chang, Dit-Yan Yeung, and Yimin Xiong. Super-resolution through neighbor embedding. In *Proceedings of the 2004 IEEE Computer Society Con-*

-
- ference on Computer Vision and Pattern Recognition, 2004. CVPR 2004.*, volume 1, pages I–I, 2004.
- [40] Chao Dong, Chen Change Loy, and Xiaoou Tang. Accelerating the super-resolution convolutional neural network. 2016.
 - [41] Jiwon Kim, Jung Kwon Lee, and Kyoung Mu Lee. Accurate image super-resolution using very deep convolutional networks. 2015.
 - [42] Christian Ledig, Lucas Theis, Ferenc Huszar, Jose Caballero, Andrew Cunningham, Alejandro Acosta, Andrew Aitken, Alykhan Tejani, Johannes Totz, Zehan Wang, and Wenzhe Shi. Photo-realistic single image super-resolution using a generative adversarial network. 2016.
 - [43] Xintao Wang, Ke Yu, Shixiang Wu, Jinjin Gu, Yihao Liu, Chao Dong, Chen Change Loy, Yu Qiao, and Xiaoou Tang. Esrgan: Enhanced super-resolution generative adversarial networks. 2018.
 - [44] Jingyun Liang, Jiezhong Cao, Guolei Sun, Kai Zhang, Luc Van Gool, and Radu Timofte. Swinir: Image restoration using swin transformer. 2021.
 - [45] Antoine Doury, Samuel Somot, Sebastien Gadat, Aurélien Ribes, and Lola Corre. Regional climate model emulator based on deep learning: concept and first evaluation of a novel hybrid downscaling approach. *Climate Dynamics*, 60(5–6):1751–1779, July 2022.
 - [46] Neelesh Rampal, Sanaa Hobeichi, Peter B. Gibson, Jorge Baño-Medina, Gab Abramowitz, Tom Beucler, Jose González-Abad, William Chapman, Paula Harder, and José Manuel Gutiérrez. Enhancing regional climate downscaling through advances in machine learning. *Artificial Intelligence for the Earth Systems*, 3(2), April 2024.
 - [47] Jorge Baño-Medina, Maialen Iturbide, Jesús Fernández, and José Manuel Gutiérrez. Transferability and explainability of deep learning emulators for regional climate model projections: Perspectives for future applications. *Artificial Intelligence for the Earth Systems*, 3(4), October 2024.
 - [48] Julien Boé, Alexandre Mass, and Juliette Deman. A simple hybrid statistical–dynamical downscaling method for emulating regional climate models over
-

- western europe. evaluation, application, and role of added value? *Climate Dynamics*, 61(1–2):271–294, November 2022.
- [49] Chia-Hao Chiang, Zheng-Han Huang, Liwen Liu, Hsin-Chien Liang, Yi-Chi Wang, Wan-Ling Tseng, Chao Wang, Che-Ta Chen, and Ko-Chih Wang. Climate downscaling: A deep-learning based super-resolution model of precipitation data with attention block and skip connections. 2024.
- [50] Fabio Merizzi, Andrea Asperti, and Stefano Colamonaco. Wind speed super-resolution and validation: from era5 to cerra via diffusion models. 2024.
- [51] Elena Tomasi, Gabriele Franch, and Marco Cristoforetti. Can ai be enabled to perform dynamical downscaling? a latent diffusion model to mimic kilometer-scale cosmo5.0_{clm9}simulations. *GeoscientificModelDevelopment*, 18(6) : 2051–2078, April2025.
- [52] Shailesh Kumar Jha, Vivek Gupta, Priyank J. Sharma, Anurag Mishra, and Saksham Joshi. Deep learning super-resolution for temperature data downscaling: a comprehensive study using residual networks. *Frontiers in Climate*, 7, May 2025.
- [53] D. E. Rumelhart, G. E. Hinton, and R. J. Williams. *Learning internal representations by error propagation*, page 318–362. MIT Press, Cambridge, MA, USA, 1986.
- [54] Francisco J. Martinez-Murcia, Andres Ortiz, Juan-Manuel Gorriz, Javier Ramirez, and Diego Castillo-Barnes. Studying the manifold structure of alzheimer’s disease: A deep learning approach using convolutional autoencoders. *IEEE Journal of Biomedical and Health Informatics*, 24(1):17–26, 2020.
- [55] Ben Garber, Aitan Grossman, and Sonja Johnson-Yu. Image super-resolution via a convolutional neural network. 2020.
- [56] Generalizing the enhanced-deep-super-resolution neural network to brain mr images: A retrospective study on the cam-can dataset. 11:ENEURO.0458–22.2023, May 2024.
- [57] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. 2015.

- [58] Laura Conangla, Joan Cuxart, Maria Antonia Jiménez, Daniel Martínez-Villagrasa, Josep Ramon Miro, Davide Tabarelli, and Dino Zardi. Cold-air pool evolution in a wide pyrenean valley. *International Journal of Climatology*, 2018, vol. 38, num. 6, p. 2852-2865, 2018.
- [59] Yuhao Gong, Yuchen Zhang, Fei Wang, and Chi-Han Lee. Deep learning for weather forecasting: A cnn-lstm hybrid model for predicting historical temperature data. *arXiv preprint arXiv:2410.14963*, 2024.
- [60] Cort J Willmott and Kenji Matsuura. Advantages of the mean absolute error (mae) over the root mean square error (rmse) in assessing average model performance. *Climate research*, 30(1):79–82, 2005.
- [61] Cort J Willmott and Kenji Matsuura. On the use of dimensioned measures of error to evaluate the performance of spatial interpolators. *International journal of geographical information science*, 20(1):89–102, 2006.
- [62] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. 2014.
- [63] Martín Abadi et al. TensorFlow: Large-scale machine learning on heterogeneous systems, 2015. Software available from tensorflow.org.
- [64] Pedregosa et al. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [65] J. D. Hunter. Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007.
- [66] Hua Li, Jing Lu, Helong Tong, and Yijun Liu. Impact of high temperature heat waves on ocean carbon sinks: Based on literature analysis perspective. *Journal of Sea Research*, 198:102487, 2024.
- [67] Haris Iqbal. Harisiqbal88/plotneuralnet v1.0.0, December 2018.
- [68] Wenming Yang, Xuechen Zhang, Yapeng Tian, Wei Wang, Jing-Hao Xue, and Qingmin Liao. Deep learning for single image super-resolution: A brief review. *IEEE Transactions on Multimedia*, 21(12):3106–3121, 2019.

- [69] Arturo A. Keller, Kendra L. Garner, Nalini Rao, Eladio Knipping, and Jeffrey Thomas. Downscaling approaches of climate change projections for watershed modeling: Review of theoretical and practical considerations. *PLOS Water*, 1(9):e0000046, September 2022.